Date of publication xxxx 00, 0000, date of current version xxxx 00, 0000.

Digital Object Identifier 10.1109/ACCESS.2017.DOI

Smart Fault Diagnostics: A Novel Hybrid AI-Driven Approach for Fault Detection in Induction Motors Under Variable Speed Operations

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ABSTRACT Faults in an induction motor (IM) can leads to an unexpected downtime, resulting in considerable economic and productivity losses. From existing literature, conventional fault diagnosis approaches in an IM struggle to reliably identify fault patterns at different speeds, particularly under changing load conditions. To resolve this issue, this paper presents a unique hybrid CNN-LSTM topology for diagnosing faulty patterns in an IM under loaded and unloaded variable speed settings. The proposed method can identify important flaws such as rotor imbalances, misalignment, stator winding issues, voltage imbalances, broken rotor bars, and broken bearings. Experiments performed using UOEMD-VAFCVS dataset reveal that the accelerometer is 99.93% accurate at constant speed and 99.96% at variable speed in both loaded and unloaded conditions. In terms of fault diagnostic accuracy in an IM operating at different speeds, this methodology outperforms cutting-edge methodologies in the literature. Moreover, using the publicly available CWRU dataset, this study validate the robustness of the proposed methodology in terms of operational issues in an IM. Finally, the proposed method achieves incredible results at varying speeds, stressing the need of improving industrial equipment reliability and maintenance methods.

INDEX TERMS Induction motor, Fault diagnosis, Acoustic signals, Vibrational signals, CNN-LSTM, Electro-Mechanical faults

Abbreviations

I. INTRODUCTION

Electric motors are the cornerstone of all manufacturing and processing industries [1]. Induction Motors (IM) are frequently used in industrial, residential, and commercial applications due to their low cost, reliability, and simplicity [2], [3]. Due to their low maintenance cost, they become the vital part of large machinery used in the industry [4], [5]. However, they can exhibit different fault conditions including electrical and mechanical like rotor imbalances, misalignments, problems with the stator windings, voltage imbalances, broken rotor bars, and broken bearings under different speed and load condition [6]. One of the most significant challenges in IM health monitoring lies in elucidating the complex mapping relationship between motor features and failure modes, which becomes more challenging in case of speed fluctuations that arises from start-stop cycles and varying load conditions. Therefore, a trustworthy prognosis methodology is essentially required to ensure the proper functioning of Induction motors under dynamic speed conditions.

The computed order tracking (COT) and its derivatives remain active research areas for fault diagnosis in Induction motors under constant and variable speed conditions. Nevertheless, the computational results are not highly precise, and these procedures necessitate expensive apparatus and highly skilled personnel to mon-

TABLE 1. List of Abbreviations

Abbreviation	Full Form
ANN	Artificial Neural Network
BiLSTM	Bidirectional Long Short-Term Memory
CEEMDAN	Complete Ensemble Empirical Mode Decomposition with Adaptive Noise
CNN	Convolutional Neural Network
COT	Computed Order Tracking
CWRU	Case Western Reserve University
CWT	Continuous Wavelet Transform
DAE	Denoising Autoencoder
DAMGCN	Domain-Adversarial Multi-Graph Convolutional Network
DBN	Deep Belief Network
DCNN	Dimensional Convolutional Neural Network
DFT	Discrete Fourier Transform
DL	Deep Learning
DWT	Discrete Wavelet Transform
FFT	Fast Fourier Transform
HHT	Hilbert-Huang Transform
IM	Induction Motor
IMS	Inductive Monitoring System
LSTM	Long Short-Term Memory
MCNN	Multi-Channel Convolutional Neural Network
ML	Machine Learning
PCA	Principal Component Analysis
PSO	Particle Swarm Optimization
RNN	Recurrent Neural Network
RP	Recurrence Plot
SDP	Symmetrized Dot Pattern
SL	Supervised Learning
SVM	Support Vector Machine
TL	Transfer Learning
UOEMD-VAFCVS	University of Ottawa Electric Motor Dataset - Variable Amplitude Fault Condition Vibration Signals
VMD	Variational Mode Decomposition
VSF-ST	Variable Scale Feature-Synchro squeezing Transform
WST	Wavelet Scattering Transform
WT	Wavelet Transform
XGB	XGBoost - Extreme Gradient Boosting

itor the precise root cause of the malfunction incident. To address this issue, a variety of advanced signal processing techniques has been reported in the literature, including the Fast Fourier Transform (FFT), Discrete Wavelet Transform (DWT), Discrete Fourier Transform (DFT), and Hilbert-Huang Transform (HHT), which can effectively analyze the signals of healthy and faulty motors for fault detection. However, these methods can provide only specific information based on the analyzed and acquired vibrational signals. Therefore, different signal decomposition methods, such as variational mode decomposition [7], empirical Wavelet Transform(WT) [8], chirp mode decomposition [9], generalized dispersive mode decomposition [10], and central frequency mode decomposition [11] are also developed to detect the faults in the motors in less time with high accuracy.The performance of these methods remains dependent on different parameters and can provide specific information based on the tunable parameters.

The recent trend in the existing literature shift towards an emerging machine learning (ML) technique. Firstly, early ML models that utilized principle component analysis (PCA) and support vector machine (SVM) succeeded significantly in fault diagnosis [12]. However, these models required manual feature extraction, which is time-consuming and expert-dependent. Therefore, deep learning (DL) has recently captured the attention of scientific experts, which provides an optimized result, especially in mechanical fault feature recognition and bearing failures under constant and varying speed conditions in less time with high accuracy. The next phase of DL is the advent of deep learning models like deep belief network (DBN), artificial neural network (ANN), recurrent neural network (RNN) and convolutional neural network (CNNs). Among these, CNNs provide a more optimized results through accurately diagnosing faults in an IM through automating feature extraction, thus improving the diagnostic precision and accuracy. For example, in [13], one dimensional convolutional neural network (DCNN) are used on case western reserve university (CWRU) and inductive monitoring system (IMS) dataset to detect and diagnose the faults of IM. The reported model used the data of mechanical faults especially in bearings of an IM operating at constant speed and an achieved accuracy up to 93.3%. However, this approach requires a large dataset and high computation power to get the desired results. To further extend the topology in [13], Shao et al. in [14] introduced deep convolutional neural network (DCNN) on vibration and current signals for fault diagnosis under six different conditions of IM and achieved the 99.8% accuracy under constant speed of IM. One of the main limitations of this approach is that this model requires data augmentation and hyper-parameter tuning issues.

To resolve this issue,a hybrid symmetrized dot pattern (SDP)- DCNN model is presented in [15] to diagnose the faults of motor under variable speed by applying angular domain resampling and complete ensemble empirical mode decomposition with adaptive noise (CEEMDAN) approaches, thus achieve 97.14% accuracy of fault diagnosis. The limitation of this approach is poor diagnosis accuracy for outer ring crack of bearing of the motor. Similarly, in [16], authors developed a multichannel DCNN classifier using multi-axis vibration data, thus achieving an enhanced accuracy compared to single-axis methods by using IMS dataset of IM for constant speed faults. However, this accuracy is compromised in case of variable speed operations. A variational mode decomposition (VMD)-DenseNet hybrid techniques is reported in [17] to convert 1D signals

to 2D time-frequency graphs for variable speed faults by using university of Ottawa IM dataset. This model required parameter optimization and achieved 92% fault diagnoses accuracy. Similarly, in [18], authors proposed a semi-supervised model using residual signals, eliminating the need for labeled data and feature engineering by applying denoising autoencoder (DAE) and CNN techniques, thus achieve an accuracy of 99.6%. However, this approach requires further optimization of DAE and CNN architectures and lacks hyperparameter tuning.In [19], variable scale feature - synchrosqueezing transform (VSF-ST) method for vibration-speed fusion with self-attention and sparse autoencoder to classify faults and achieve the accuracy of 95.18% for Xi'an Jiao tong University dataset under variable speed of IM. Moreover, in [18], wavelet scattering transform (WST) - XGBoost - extreme gradient boosting (XGB) - artificial neural network (ANN) hybrid model is presented to simplifying bearing classification and resolving signal-based feature extraction complexity and achieve 99.5% fault diagnosis accuracy for variable speed for limited and less severe fault conditions. Furthermore, hybrid model is reported in [20] which the combination of feature-rich 2D signal transformation with enhanced noise immunity and generalization and achieve 97.8% accuracy. However,it required very high computation and recurrence plot (RP) image coding cost for variable speed dataset.

To enhance the cross-domain, feature extraction and classification a domain-adversarial multi-graph convolutional network (DAMGCN) is reported in [21] which achieved 99.4% accuracy for CWRU dataset under variable speed condition. However, one of the major drawbacks is that this method requires a complex model integration and address only bearing related mechanical faults. Similarly, Suawa et al. in [22] present a novel topology of sensor fusion in terms of an extension of previous works, which significantly improved the precision of fault identification and achieved an accuracy of 98.8% for DCNN, 93.5% for CNN-LSTM, and 73.6% for LSTM. Moreover, in [23], authors presented wavelet scattering transform-based hybrid CNN-SVM optimized technique for real-time industrial fault diagnosis under dynamic conditions. The reported method achieved accuracy of 100% which will be compromised for diagnosisng multiple faults at the same time.An improved version is reported in [24] which utilized an image-based DL method with fine-tuned transfer learning networks using IMFs and SDP images authors proposed an image-based DL method using IMFs and SDP images. However, nonlinearity and non-stationarity of vibration signals are not addressed in this method and achieve 99% accuracy for CWRU dataset of variable speed. To further extend it, a hybrid algorithm combining REMD for feature extraction, EO for feature selection, and RF, ET, and KNN for fault classification is reported is reported in [25].The reported approach improved the gearbox fault diagnosis and achieved 99.24% accuracy for variable speed of gearbox.

A recent advanced techniques in [1] utilized the combination of CNN with continuous wavelet transform (CNN-CWT) for constant speed fault detection and achieve accuracy of 97%. Similarly, in [26], authors presented multiscale fusion residual network and multiple winning self-consciousness, self-organizing map for motor fault detection and achieve high accuracy in noisy environment. However, still this model required more parameters and computations than simple CNN models and relies on supervised learning (SL) which limits its scalability in industrial settings with limited labeled samples. To further extend the above-mentioned topologies, hybrid CNN- bidirectional long short-term memory (BiLSTM) with particle swarm optimization (PSO) is reported in [27] for bearing fault diagnosing under variable speed of the motors. The reported model reduce parameter tuning costs and enable transfer learning (TL) to new conditions. However, the limited training data under diverse working conditions along with other challenges in parameter tuning is unable to justify the accuracy 99.14% of this model. In [28], an multi-channel convolutional neural network (MCNN)-BiLSTM model is presented with channel attention, utilizing multi-scale convolution for better feature extraction and achieving improved accuracy of 99.68% by using CWRU dataset. However, this model only addresses the bearing related faults of IM. To resolve this issue, a hybrid CNN-LSTM model is reported in [6], which provides high accuracy and addresses both mechanical and electrical faults of motor by using a new comprehensive dataset of motors having seven different mechanical and electrical faults of IM and achieve 99.96% accuracy. However, the reported model also fails to address the faults under variable speed of the motors. The overall comparison of above discussed different ML model and their performance is given in Table. 1.

IM faults under Variable load and speed conditions have been investigated in several different research studies. In [29] authors proposed an integrated approach by combining Continuous Wavelet Transform (CWT) with LeNet-5 and Long Short-Term Memory (LSTM) for the detections of mechanical faults under variable load conditions. In this model CNN employed to extract complex features while LSTM networks to capture temporal dependencies in vibration signals. This model achieved high accuracy ranging from 92.5% to 99.6% under variable load conditions for four different sensors. However, the primary focuses of this work is to diagnose the only bearing mechanical faults under variable load conditions while maintaining a constant speed. The comprehensive analysis of the effects of both variable speed and varying load for both electrical and mechanical

faults in a combined framework is not addressed in this study. Additionally, the unloaded operating conditions are not explicitly considered in this work, which are common in real-world industrial applications. In [30] authors presented a hybrid DCNN-LSTM approach that eliminates data preprocessing and traditional signal processing methods requirements for feature extractions. The model effectively classifies faults under variable speed conditions and different fault depths while achieving high accuracy (99.40% for variable speed datasets and 99.87% for diverse fault depth datasets). However, this study only addresses the bearing faults and does not address the broader spectrum of both mechanical and electrical faults that commonly encountered in industrial motors. Additionally, the model performance and accuracy are high under variable speeds, it has not been explicitly evaluated its robustness across varying load conditions and both loaded and unloaded states simultaneously have not been considered.

In contrast, our proposed approach presents a comprehensive fault diagnosis framework to addresses these limitations by considering both variable and constant speed and varying load conditions, as well as both loaded and unloaded scenarios. Unlike the previous works, seven different diverse faults of Induction motors, including both mechanical (e.g., rotor unbalance, bowed rotor, faulty bearings) and electrical (e.g., voltage unbalance, single phasing, stator winding fault) failures are evaluated in our research work. We use the UOEMD-VAFCVS dataset, which provides extensive coverage of real-world operating conditions, making our model more applicable to industrial settings. Our CNN-LSTM architecture effectively captures both spatial and temporal features, leading to superior classification performance. Our model achieves 99.93% accuracy for constant speed and 99.96% accuracy for variable speed conditions, demonstrating its robustness across different operating scenarios. Furthermore, we validate our approach on the CWRU dataset, where our model significantly outperforms existing techniques, reinforcing its generalizability and practical applicability. The comparison of previous works and our proposed work is given in Figure.1 highlighting that our model is successfully addresses both electrical mechanical faults under variable speed and load conditions. While all other previous works did not address these challenges simultaneously.

Although the above-mentioned algorithms have achieved satisfactory results in large machinery and equipment applications, there are still thorny issues that need to be addressed:

1) Both electrical and mechanical faults of an IM at constant and variable speed is still one of the challenging research issue in the existing literature.
2) Under variable load and speed condition, the amplitude of vibrational signals behaves differently as compared to constant speed, which is also one of the main limitations in the existing literature, that need to be addressed.
3) With CNN alone, it is difficult to obtain feature information of faulty conditions of motor under variable speed and load conditions as amplitude of received signals is continuously varying.
4) Finally, fault detection by CNN becomes more challenging for both constant and variable speed conditions simultaneously as fault pattern/feature information varies abruptly

In order to address the current limitations, a new hybrid ML and deep learning model CNN-LSTM is proposed in this work. Firstly, segmentation is performed on the data coming from different sensors to differentiate the healthy and faulty patterns of motors for dynamic load and speed condition. Secondly, the CNN layer is used to extract the feature information of motor health conditions. Finally, LSTM is used to obtain fault feature from CNN, as the available data is time-dependent, and the acquired features are then applied in Softmax classifier for motor fault classification. The proposed hybrid CNN-LSTM model is validated by two different publicly available dataset of IM including university of ottawa electric motor dataset – vibration and acoustic faults under constant and variable speed conditions (UOEMD-VAFCVS) and CWRU.

The novelty of our hybrid CNN-LSTM model is that it can solve the challenges faced by varying speed and variable load conditions, which are one of the main limitations in the existing literature. The proposed model can diagnose both electrical and mechanical faults in an induction motor, thus outperforming the existing studies, which only diagnosis the mechanical faults under varying speed or varying load conditions. The superior performance of the proposed model is obtained with the use of the dataset, which contains motor health data under both constant and variable speeds and loaded and unloaded conditions and setting up the optimal values of hyper parameters for both CNN and LSTM after several experiments and extensive grid search. Thus, making the proposed model for better feature extraction and temporal dependencies, and making it generalized well across several types of faults and variations.

To summarize, the major contributions of our work are as follows.

1) Data from different sensors is segmented to classify the different fault patterns at the same time.
2) The proposed hybrid CNN-LSTM successfully classifies the Induction motors different faults with an accuracy of 99.96%, including electrical and mechanical for different speeds and load condition on CWRU and UOEMD-VAFCVS dataset.
3) The consistency of the proposed model is validated through several runs and k-fold validation.

TABLE 1. Comparison of different ML model for Motor fault classification

Reference	Year	Problem Formulation	Model	Dataset	Constant /Variable speed	Accuracy (%)	Limitation	Contribution
[12]	2015	Combining permutation entropy, ensemble empirical mode decomposition for fault identification.	PCA, SVM	CWRU	Constant Speed	90%	Manual Feature extraction.	Low computational cost and Fast.
[13]	2019	An efficient adaptive 1D CNN classification for generic and intelligent bearing fault diagnosis.	CNN	CWRU and IMS	Constant and variable	93.9%	Requires large datasets and computational resources, Less interpretable than traditional methods.	Excellent at feature extraction from raw data, High performance in image and signal processing.
[14]	2019	Method serves as approach for IM fault diagnosis.	DCNN	Vibration and current signals under 6 conditions	Constant speed	99.8%	Requires data augmentation and further hyperparameter tuning.	Developed a multi-signal fault diagnosis framework using CNN and TFDs for robust and automatic feature extraction.
[15]	2020	An effective bearing fault diagnostic using angular domain resampling combined for managing diverse operating conditions.	SDP and DCNN	-	Variable speed	97.14%	Poor diagnosis accuracy for outer ring crack.	Proposed a method using angular domain resampling, CEEMDAN for fault diagnosis under varying conditions.
[16]	2021	A multichannel, multilevel classifier to enhance detected bearing faults.	1D-CNN	IMS	Constant speed	100%	Requires testing with variable-speed applications.	Developed a multichannel classifier using multi-axis vibration data, achieving enhanced accuracy compared to single-axis methods.
[17]	2021	Improves with fault diagnosis and forecasting methods for time-varying bearings that use DL algorithms.	VMD-DenseNet	University of Ottawa data of the time-varying bearing	Variable speed	92%	Parameter optimization needed, lacks comparison with traditional methods and statistical confidence intervals.	Proposed method for converting 1D signals to 2D time-frequency graphs.
[18]	2021	A framework approach to classify bearing faults in IM.	DAE and CNN	-	Constant Speed	99.6%	Requires further optimization of DAE and CNN architectures, lacks hyperparameter tuning.	Proposed a semi-supervised model using residual signals, eliminating the need for labeled data and feature engineering.
[19]	2021	A vibration-speed fusion network with self-attention and sparse feature extraction performs bearing fault diagnosis in time-varying systems.	VSF-ST	Xi'an Jiaotong University	Variable Speed	95.18%	Computational complexity due to multi head attention.	Proposed a vibration-speed fusion model with self-attention and sparse autoencoder to classify faults.
[20]	2022	Effective fault identification methods for rolling bearings under uncertain conditions.	VMD–RP –CSRN	Jiangnan University dataset	Variable Speed	97.8%	High computational cost for RP image coding.	Combination for feature-rich 2D signal transformation with enhanced noise immunity and generalization.

TABLE 1 (Continue). Comparison of different ML model for Motor fault classification

Reference	Year	Problem Formulation	Model	Dataset	Constant /Variable speed	Accuracy (%)	Limitation	Contribution
[21]	2022	A domain-adversarial multi-graph convolutional network operates for unsupervised domain adaptation in rolling bearing fault diagnosis over different operational scenarios.	DAMGCN	CWRU	Variable Speed	99.4%	Requires complex model integration.	Proposes a domain-adversarial multi-graph convolutional network for cross-domain, enhancing feature extraction and classification.
[22]	2022	The sensor data fusion method for BLDC motor modelling and fault detection.	DCNN, CNN-LSTM	Microphone and accelerometer sensors	constant speed	98.83%	Limited sensor configurations and inability to handle environmental noise.	Show superiority of DL for raw multi-sensor data analysis.
[31]	2022	A scheme for bearing diagnosis of IM by utilizing wavelet scattering transform features.	WST-XGB-ANN	public dataset for 3 bearing conditions	Variable Speed	99.5%	Does not address fault severity analysis.	Simplifying bearing classification and resolving signal-based feature extraction complexity.
[23]	2023	A real-time robust bearing fault detection method that uses scattergrams with a hybrid NN structure.	CNN-SVM	Lab-based data	Variable Speed	100%	Higher cost for sound sensors and no mixed fault diagnosis.	Wavelet Scattering Transform based hybrid optimized for real-time industrial fault diagnosis under dynamic conditions.
[24]	2023	Image-based DL networks for fault diagnosis use vibration signals, which create data image representations to improve fault detection.	Alexnet, VGG, ResNet	CWRU	Variable speed	99%	Challenges in nonlinearity and non-stationarity of vibration signals.	Proposed an image-based DL method using IMFs and SDP images and implemented with fine-tuned transfer learning networks.
[25]	2023	A gearbox fault diagnosis with REMD and EO techniques that worked with ML classifiers to boost fault detection capabilities.	Machine Learning Classifiers	Real-time vibration signals from a gearbox	Variable speed	99.24%	Limited to vibration signals and requires hybrid data acquisition procedure with multiple sensors for more accurate results.	Developed a hybrid algorithm combining REMD for feature extraction, EO for feature selection, and RF, ET, and KNN for fault classification, improving gearbox fault diagnosis accuracy.
[1]	2024	CWT is employed in combination with CNN to diagnose bearing faults in IM.	CNN with CWT	MFPT	constant speed	97%	-	Proposed approach with Scalograms, presents a promising solution for real-world applications.
[26]	2024	An advanced competitive multi-scale network system for rotary machinery fault diagnosis to enhance detection capabilities regardless of operating conditions.	IMSFFRN-MCSOM-SVM	CWRU, PU, SEU, gear	Both	100%	Requires more parameters and computations than simple CNN models, and relies on supervised learning which limits its scalability in industrial settings with limited labeled samples.	Achieving high accuracy and robustness in noisy environments.

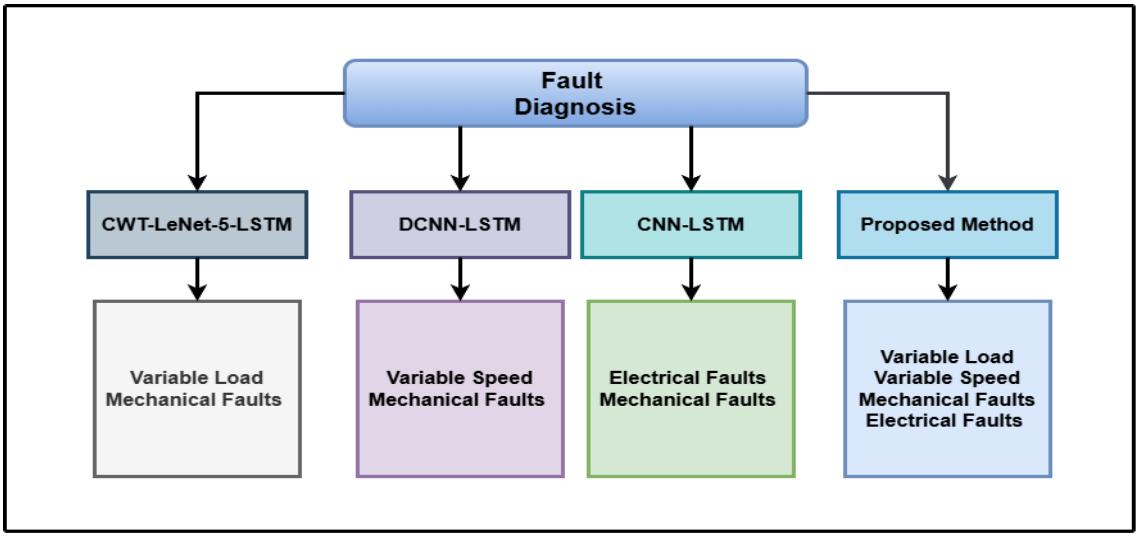

FIGURE 1. Flow chart explain the previous work and proposed work comparison

TABLE 1 (Continue). Comparison of different ML model for Motor fault classification

Reference	Year	Problem Formulation	Model	Dataset	Constant /Variable speed	Accuracy (%)	Limitation	Contribution
[27]	2024	Efficient design addresses the training sample scarcity for fault detection in bearings while operating under various working conditions.	CNN-BiLSTM with PSO	Dataset named as A, B, C, and D	Variable speed	99.14%	Limited training data under diverse working conditions and challenges in parameter tuning.	Reduce parameter tuning costs, and enable transfer learning to new conditions.
[28]	2024	Rolling bearing fault diagnosis for improved diagnostic performance through feature extraction and attention-based mechanisms.	MCNN-BiLSTM	CWRU	Variable speed	99.68%	-	Proposed model with channel attention, utilizing multi-scale convolution for better feature extraction and achieving improved accuracy.
[6]	2024	DL framework detects multiple faults in IM through diverse sensor data to enhance system reliability and fault recognition accuracy.	CNN-LSTM	UOEMD-VAFCVS	Constant speed	99.96%	Unable to diagnose the faults under variable speed condition.	Provides good accuracy and address both mechanical and electrical faults of Motor.
-	2025	Our Proposed	CNN-LSTM	UOEMD-VAFCVS	Both Variable and Constant speed	99.96%	Unable to diagnose the faults under variable speed condition.	Provides good accuracy and address both mechanical and electrical faults of Motor.

4) An extensive comparison with recent state of the art work is presented in terms of accuracy, F1 score computational cost.
5) The proposed hybrid CNN-LSTM successfully classifies and identifies the IM different faults with an accuracy of 99.96%, including electrical and mechanical for different speeds and load condition.
6) Finally, the accuracy and performance of the proposed model are evaluated by using two different (CWRU and UOEMD-VAFCVS) dataset.

The rest of the article is organized as follows. Section II briefly introduces the CNN-LSTM framework. Section III elaborates methodology and effectiveness of CNN-LSTM for UOEMD-VAFCVS and CWRU dataset. Finally, section IV concludes the article with future recommendations.

II. FAULTS CHARACTERISTICS AND IMPACT ON IM

Every fault has its own unique properties, and machine's vibrational signals and electrical behavior vary due to each fault, which influences the model in diagnosing and classifying it. We have addressed seven different faults including five mechanical and two electrical faults in this study. Details description of each fault and its impact on the machine is given below:

Rotor Unbalance

An unbalanced rotor is a rotor having mass distributed unevenly about the axis of rotation, which causes excessive centrifugal forces. This imbalance causes significant radial vibration, increasing stress in the motor bearings and housing. The leading causes are manufacturing defects, wear and tear, or improper installation. The vibration spectrum generally contains a major frequency component at the running speed (1x RPM) and harmonics. Rotor unbalance that is not attended to can lead to increased bearing wear and secondary faults [32].

Rotor Misalignment

When two or more axes on a shaft/rotor are not aligned properly, it leads to a transfer of uneven force, resulting in increased torsional vibrations. This misalignment is often angular or parallel, which increases friction and thus brings forward the wear of bearings and couplings. The motor was incorrectly installed, or mechanical wear is a common cause. In general, it contains the fundamentals of the spectral signature and sideband frequency components around the fundamental frequency, which are obscured under variable speed conditions, leading to complex detection of the fault [33].

Stator Winding Fault

Electrical faults due to insulation breakdown, turn-to-turn short circuits, or thermal degradation are stator winding faults. This results in unbalanced phase currents, high temperatures, and poor motor efficiency. The fault significantly changes the magnetic flux distribution and can lead to complete motor failure if it is not detected in the early stage. A stator winding fault gives rise to characteristic harmonic frequencies (e.g., 2 times line frequency) in the current spectrum, which electrical analysis techniques can diagnose [34].

Voltage Unbalance Single Phasing

It is called voltage unbalance when phase voltages are not equal or when one phase is wholly lost. This results in higher negative sequence currents, heating, and pulsating torque. Common causes include faulty circuit breakers, poor electrical connections, or issues with the power supply. While single phasing is more serious and pushing, as the motor is an alternative to working in two-phase, the motor current spectrum changes dramatically, and the motor may even be shut down [35].

Bowed Rotor

A bowed rotor is a mechanical defect in which the rotor shaft has an imperceptible bend, causing the air gap between the rotor and stator to be non-uniform around the rotor. This creates irregular magnetic interactions, resulting in torque output fluctuations and motor instability. It is caused by thermal expansion, mechanical stress, or improper storage. The spectral signature displays low-frequency components in the vibration spectrum that are difficult to detect without means of advanced diagnostic techniques [36].

Broken Rotor Bars

High-load motors often suffer from mechanical stress, thermal fatigue, or defects in manufacturing, which can lead to broken rotor bars. Rotor Type Fault: When rotor bars get short-circuited and break, the current flow through the rotor is interrupted, causing low torque, high slip, and the generation of a large amount of heat. The fault of this type is not detected as a characteristic spike in the current spectrum. Instead, they appear as sidebands around the supply frequency, meaning a proper diagnosis needs to be performed using, for example, motor current signature analysis MCSA or other high-end signal processing techniques. When multiple bars break, not only does this disrupt the entire mechanical system, but the performance of the motor also dropsdrastically, which may lead to complete failure [37].

Faulty Bearings

Bearings support the rotor and aid in its rotation; however, inner race faults, outer race faults, rolling element defects, and lubrication failures result in excessive high-frequency vibrations. The errors lead to friction, heat, and rotor instability. Early detection of faulty bearings is key to avoiding catastrophic failure, achievable by analyzing their characteristic high-frequency resonance patterns in vibration, prominently visible within discrete ball-pass frequency (BPF) components [38].

Healthy Motor

A healthy motor will show a clean frequency spectrum with no unusual peaks or harmonics. If adequately maintained and closely monitored, it could provide steady performance over an extended lifetime [39].

Vibration analysis, current monitoring, and advanced signal processing techniques enable the detection of signature spectral characteristics for each fault. Early fault detection prevents failure modes, reduces downtime, and maintains motor reliability. These insights enable the development of predictive maintenance strategies, which can significantly improve the efficiency and lifetime of induction motors employed in industrial applications.

III. METHODOLOGY

This methodology section will provide detailed information about the dataset and model used for fault prediction. The Electric Motor Dataset on Vibration and

Acoustic Faults is used to conduct detection on the operational conditions of electric motors. The study focuses on the information obtained from three accelerometers positioned to capture both speed conditions and improve the ability to find faults by pattern recognition in vibration data. A CNN-LSTM model is applied to improve fault classification accuracy, using one sensory input based on constant and continuously changing speed.

A. DATASET

Data was collected from the University of Ottawa Electric Motor Dataset Vibration and Acoustic Faults under Constant and Variable Speed Conditions (UOEMD-VAFCVS) [40]. The data was acquired via a Spectra Quest Machinery Fault and Rotor Dynamics Simulator designed for this purpose. This test setup included accelerometers, a microphone, and a variable-frequency motor. These components collaborated to capture data on vibration, sound, temperature, and rotational speed utilizing LabVIEW.The set includes many motor situations. One functions well, while seven show faults, including rotor misalignment, rotor imbalance, stator winding faults, voltage imbalance, fractured rotor bars, and poor bearings under constant and variable operational speeds. The positioning of the sensors was arranged, as seen in figure 2: To monitor vibrations resulting from stator and rotor interactions, A-1 was installed at the drive end. A-2 was positioned at the motor's midpoint to evaluate overall vibration patterns, and A-3 was installed at the non-drive end to concentrate on bearing and load-related vibrations.

The data collection process was conducted using a sampling rate of 42,000 samples per second, and each recording lasted 10 seconds. The data set records speeds under eight classes, ranging from one to eight, as shown in Table 3. The data columns are organized: accelerometer data in the first, third, and fourth columns; acoustic data in the second column; and temperature data in the fifth column. Labels for the motor's health status are encoded using a combination of two letters and two numbers, as shown in Table 2. The first and second represent the class's name, the third represents speed frequency, and the fourth represents 0 for the unloaded condition and 1 for the loaded condition.

TABLE 2. Motor Health Conditions

Condition of Motor Health	Denoted as
Bowed Rotor	B_R
Faulty Bearings	F_B
Healthy	H_H
Broken Rotor Bars	K_A
Rotor Misalignment	R_M
Rotor Unbalance	R_U
Stator Winding Fault	S_W
Voltage Unbalance and Single Phasing	V_U

TABLE 3. Motor operational parameters

Frequency	Speed	Files label
15 Hz	Constant	1
30 Hz	Constant	2
45 Hz	Constant	3
60 Hz	Constant	4
15 Hz to 45 Hz	Variable	5
30 Hz to 60 Hz	Variable	6
45 Hz to 15 Hz	Variable	7
60 Hz to 30 Hz	Variable	8

B. DATA HANDLING AND PREPROCESSING

1) Data Loading

A complete dataset of 128 CSV files from electric motors running at constant and vibration speeds with time was used. These files represent seven different fault states and the motor health statuses corresponding to conditions. After importing the entire dataset, extract the constant speed files with 1, 2, 3, or 4 files representing the third component. Similarly, extract the vibration speed files represented by 5, 6, 7, or 8 from the dataset. Each CSV file was loaded into Pandas Data Frames in the following step. These data handling procedures made data manipulation easier.

2) Data Segmentation

The dataset was organized into segments to manage the raw data's complexity efficiently. Each segment consisted of 1,000 samples organized to give enough information for accurate fault classification. This method of segmentation allowed for handling continuous time-series data from each motor test. After that, each data segment was labeled according to the type of fault it represented, and these labels were stored as numbers so that they could be used more efficiently in later fault detection.

The sole significant contribution of this research work is not data segmentation; however, data segmentation plays an important role in data processing strategy. By selecting the appropriate data sample windows, our hybrid CNN-LSTM model capable to achieve higher accuracy, reduced computational cost. we ensure that electrical and mechanical fault patterns are effectively captured and diagnosed by segmenting the raw vibration data into 1,000-sample windows.

Extensive experiments are being performed with different segments sizes including 100, 500, and 1,500 samples to select the optimal 1000 samples per segments for raw vibrational data.

It is evident from the Results shown in Table. 7 that smaller segment sizes (e.g., 100 or 500) has higher Mean Squared Error (MSE), high computation time and introduced instability due to excessive fragmentation. Larger segment sizes (e.g., 1,500) have significant decease computational complexity while offering no significant accuracy benefits.

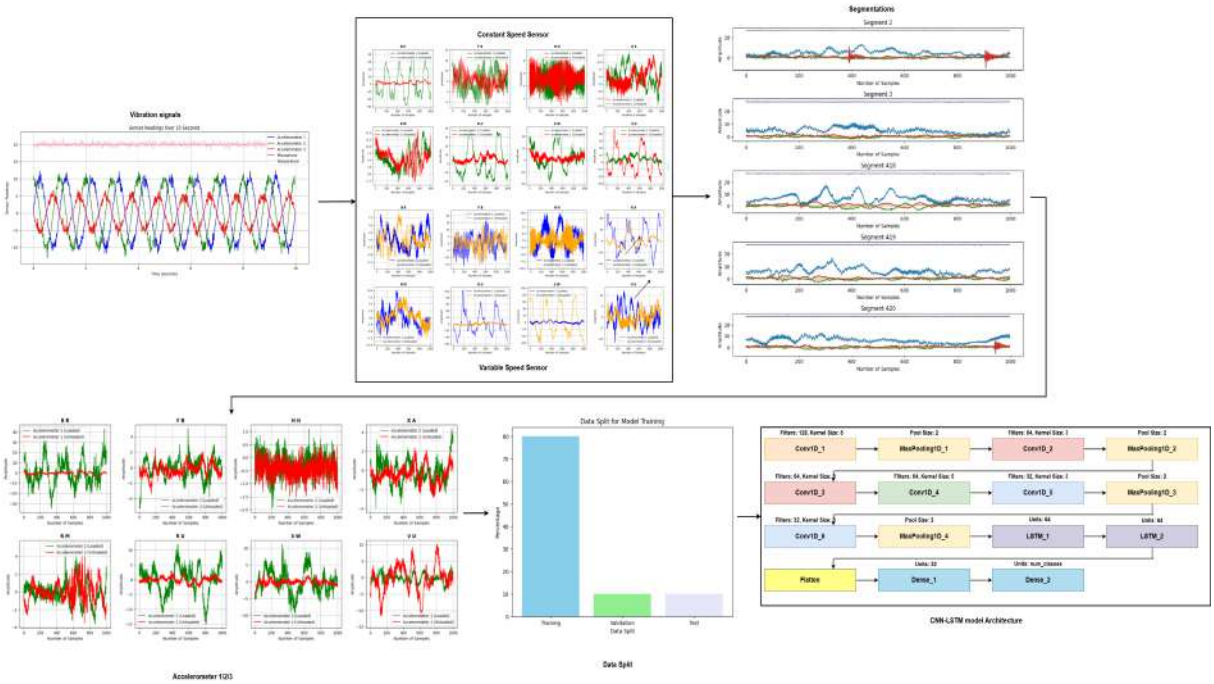

FIGURE 2. Proposed architecture for vibration signal analysis, including signal segmentation, data splitting, and model architecture

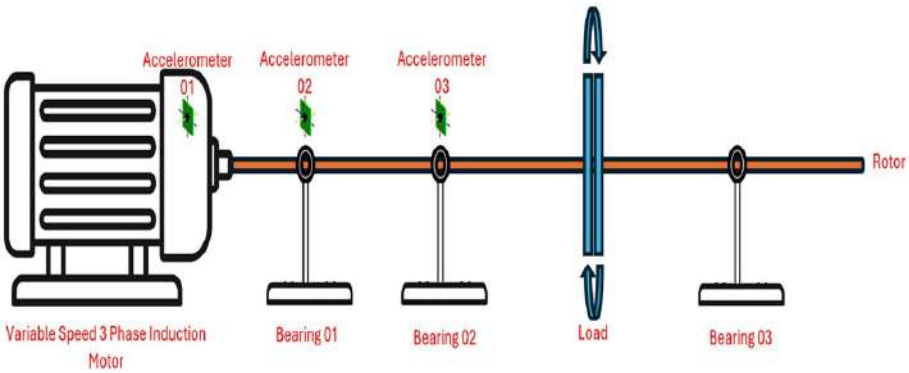

FIGURE 3. The figure illustrates the placement of accelerometers on the motor housing to capture vibrational data. Accelerometer-01 are positioned at the drive end, Accelerometer-02 at the center, and Accelerometer-03 at the non-drive end, respectively.

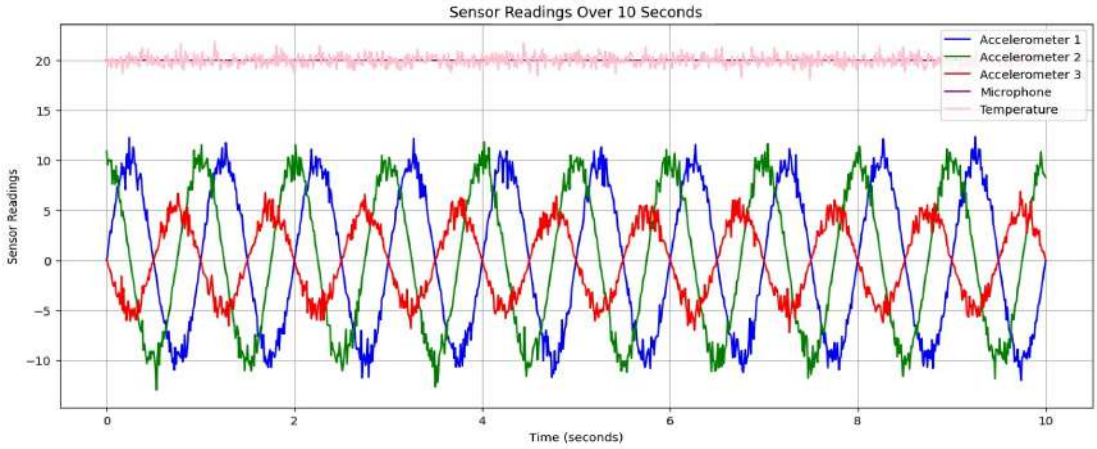

FIGURE 4. The figure illustrates the all sensor readings over a 10-second period

Visualization of segmentations on 1000 samples of seven different faults are depicted in figures below:

TABLE 7. Segment Size vs. MSE and Computational Time

Segment Size	Mean Squared Error (MSE)	Computational Time (Seconds)
100	0.005694	1977.345
500	0.00222174	715.3695
1000	0.000111666	524.34809
1500	0.0007175	515.3708

The selection of 1000 sample per segments provide the stable and computational efficient faults patterns. Due to this data preprocessing step, our hybrid CNN-LSTM model effectively utilizes this segmentation to enhance fault detection under varying load and speed conditions for diverse (electrical and mechanical) faults patterns.

3) Feature Selection and Extraction

In this phase of the study, data was gathered from each of the three accelerometers independently with time. The process began with data from Accelerometer-1 (A-1). This approach allowed for a detailed examination of each accelerometer with time contributions to fault detection, enhancing the specificity and accuracy of the model. By analyzing the data from each accelerometer separately, the unique features of the faults identified by each sensor were kept and investigated accurately.

4) Training Process

The data had to be divided into three sets, with 10% for testing, 10% for validation, and 80% for training. It was essential to ensure that the models' changes were correct and did not overfit the training data, which was done with the help of the validation set. The training set was used to teach the models how to find faults, and the test set checked how well the models worked because it gave a fair picture of how well they could apply to data.

C. MODEL DEVELOPMENT

1) Hybrid CNN-LSTM

CNNs are excellent for identifying local patterns in the data. In the context of motor fault identification, motor fault data such as vibration signals from accelerometers often have spatial (or frequency) and temporal correlation [41]. CNNs can efficiently capture these local patterns in raw time-series data by using the following mathematical equations.

CNN is applied to input data X, which has different features over different time intervals. The output of the convolutional layer is:

$$Y_{i,j} = \sigma\left(\sum_{k=1}^{m} W_k X_{j+k-1} + b\right), \quad j = 1, 2, \ldots, T' \tag{1}$$

where:

- W_k= Convolutional kernel weight for the k-th filter,
- b= Bias term,
- σ= Activation function,
- m= Filter length,
- $T' = T - m + 1$; Output sequence length after convolution.

The pooling operation is applied to the output of CNN to reduce the dimensionality:

$$P_{i,j} = \text{MaxPooling}(Z_{i,j}) \tag{2}$$

The output of CNN is denoted as H_{CNN}. The output of CNN is given to the LSTM to capture the time-dependent patterns, where the current state depends on past states. The LSTM cell has three different gates: the input gate I_t, forget gate F_t, and output gate O_t [42]. The expressions of these gates based on the output of CNN ($H_{\text{CNN},t}$) at each time step t are given as:

INPUT GATE

$$I_t = \sigma(W_I h_{t-1} + U_I H_{\text{CNN},t} + b_I) \tag{3}$$

FORGET GATE

$$F_t = \sigma(W_F h_{t-1} + U_F H_{\text{CNN},t} + b_F) \tag{4}$$

OUTPUT GATE

$$O_t = \sigma(W_O h_{t-1} + U_O H_{\text{CNN},t} + b_O) \tag{5}$$

In the above equations, σ represent Sigmoid function,$H_{\text{CNN},t}$ is Input, h_{t-1} is Hidden state,W_I, U_I, b_I, $W_F, U_F, b_F, W_O, U_O, b_O$ are Weight matrices and bias terms. The internal cell state of LSTM is:

$$C_t = (F_t \odot C_{t-1}) + (I_t \odot \tanh(W_C h_{t-1} + U_C H_{\text{CNN},t} + b_C)) \tag{6}$$

The final hidden state is:

$$h_t = O_t \odot \tanh(C_t) \tag{7}$$

where $\odot$ represents element-wise multiplication.

The final hidden state is the LSTM output, denoted by H_{LSTM}. The output of LSTM is fed to fully connected layers for classification of fault :

$$Z = \sigma(W_{FC} H_{\text{LSTM}} + b_{FC}) \tag{8}$$

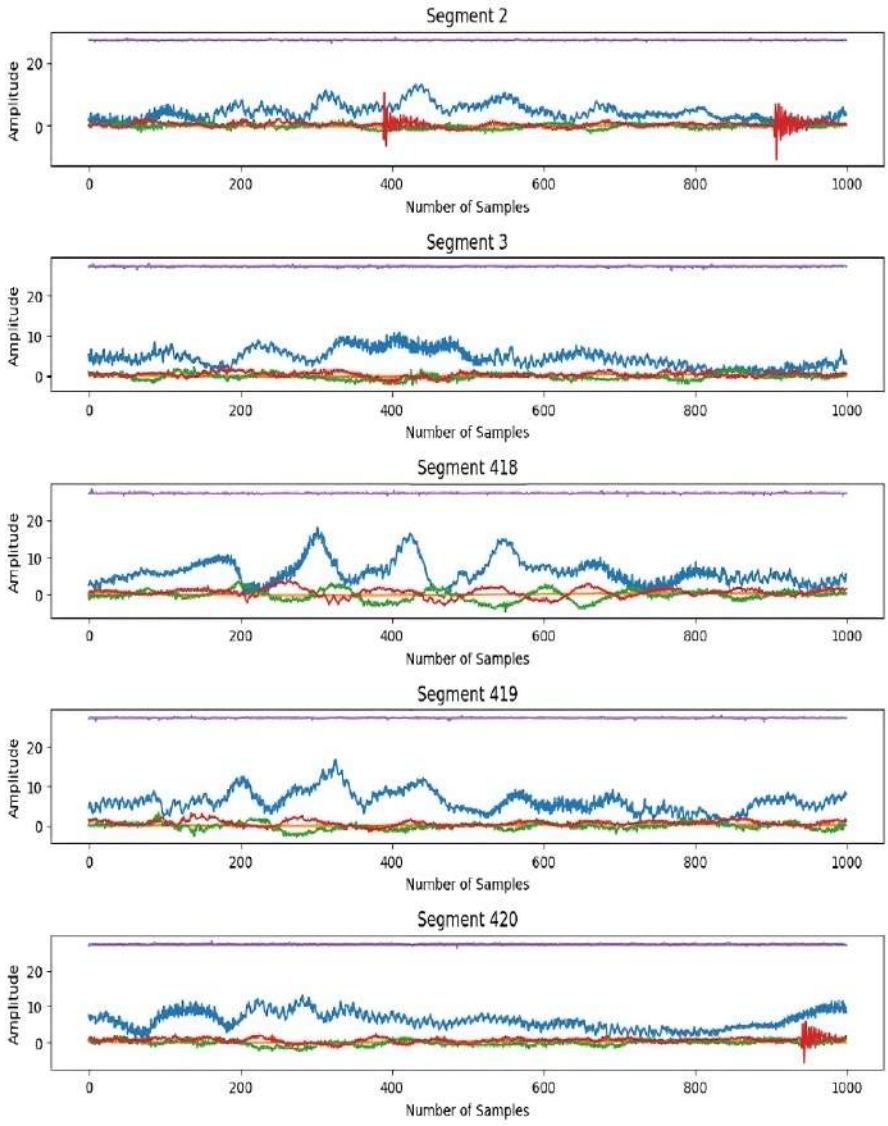

FIGURE 5. The figure illustrates the segmentation of data, where each file consists of 420,000 data points, which are divided into 420 segments, with each segment containing 1,000 samples.

W_{FC}, b_{FC} are Weights and bias term of the fully connected layer.

The final output of δ LSTM would be as follows:

$$Y_{\text{finalsim}} = \delta\Bigg(W_{\text{rel}}\Big(O_t \odot \tanh\big(f(C_{\text{et}}) \odot C_{\text{et}}\big) + f\big(O_t \odot \tanh(W_h C_{t-1} + U_{\text{man}} C)\big)\Big) + \sum_{j=1}^{m} W_j \cdot X_j^{(t)} + b_j + b_{\text{fet}}\Bigg) \tag{9}$$

This architecture shown in Figuer 5 is particularly good at recognizing complex fault pattern and changes in motor behaviour over time.

2) Model Architecture

The architecture of the CNN-LSTM model consists of several layers designed to capture the detailed characteristics of the fault data: Convolutional Layers are the initial layers of the model with 128 filters and a kernel size of 5. The ReLU activation function introduces non-linearity in this layer, which handles input data of 1000x2 with two features every time step. After convolution, a max pooling layer reduces data dimensionality, which reduces computing effort and improves dominant feature extraction by simplifying input features. Later, convolutional layers refine data features. With 64 filters

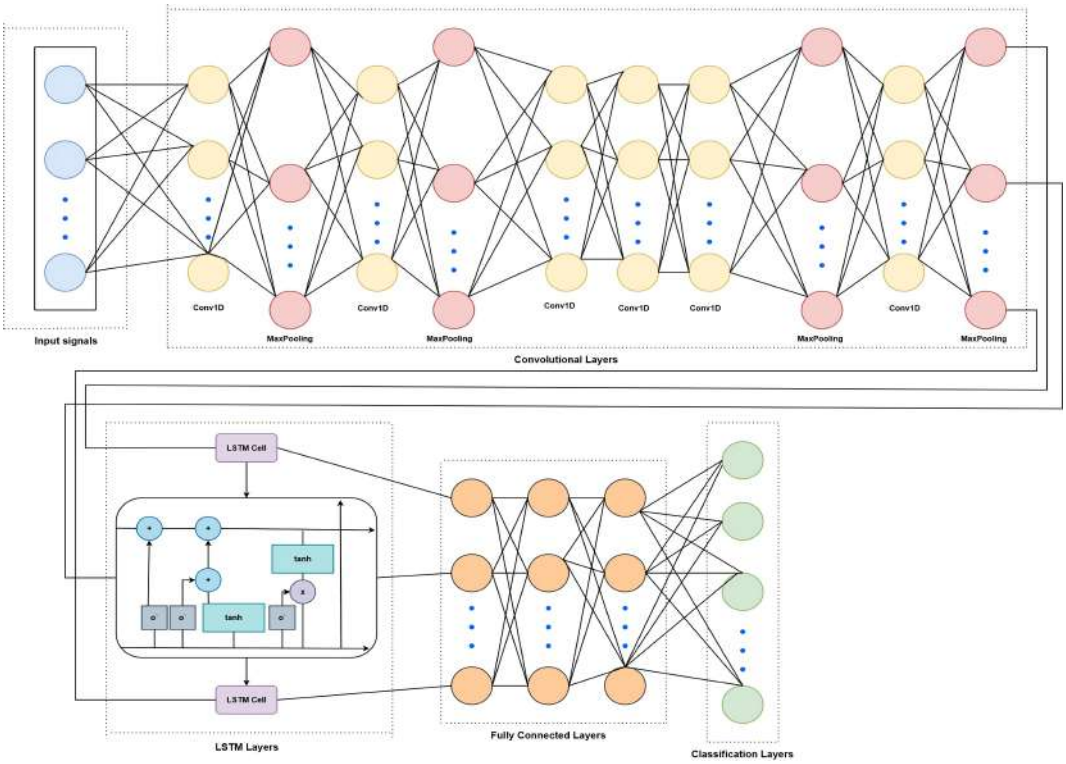

FIGURE 6. Hybrid architecture of CNN-LSTM model

and 3 to 5 kernel sizes, these layers provide a robust feature set for accurate categorization. Each convolutional layer is followed by a max pooling layer to minimize feature map spatial dimensions and improve model efficiency and generalization from training data.

The model uses LSTM layers for sequence processing after convolutional and pooling layers. The first LSTM layer has 64 units and returns sequences to preserve temporal information for learning dependencies across time steps. The next 64-unit LSTM layer summarizes temporal information into a vector representation of each sample to prepare the model for classification. Flatten and classify LSTM layer output using a dense network structure. This model has a dense layer of 32 neurons with ReLU activation and a softmax classification layer that outputs probabilities for the eight classes. This system allows accurate probabilistic fault categorization of accelerometer data.

Each training period refines model parameters to reduce the loss on a validation set, preventing overfitting and ensuring the model generalizes effectively to new data. This architecture combines CNNs' spatial feature recognition and LSTMs' temporal processing characteristics, making our model ideal for dynamic and complicated fault detection. Table 4 shows each layer of the model, including kernel type and size, units, activation functions, and data shape changes as it moves through the network.

The methodology employed in this study has proven highly effective in diagnosing various fault conditions in electric motors, such as misalignment, bearing failures, and rotor bar faults. By utilizing a meticulously designed ML model that analyzes data from multiple accelerometers under both constant and variable speed conditions, significant diagnostic successes were achieved. The inclusion of dynamic speed data has notably enhanced the accuracy of fault detection, underscoring the critical role that speed variations play in the comprehensive assessment of motor health. This approach not only improves the reliability of fault diagnostics but also contributes to the development of more robust predictive maintenance strategies in industrial applications.

IV. RESULTS & DISCUSSION

This study used the Adam optimizer with a learning rate of 0.001. The exponential decay rates for the first defined as = 0.9 and second moment estimations have been defined as = 0.999, respectively. The performed loss function was categorical cross-entropy, with a batch size of 128. From Figures 13 to 29, you can see the CNN-LSTM model's accuracy and loss graphs. These were made using data from all three accelerometers. The accuracy graphs show an ongoing and constant rise, indicating stable training and the model's capacity for effective learning. The validation accuracy curve mirrors the training accuracy, demonstrating the model's effective generalization without overfitting as

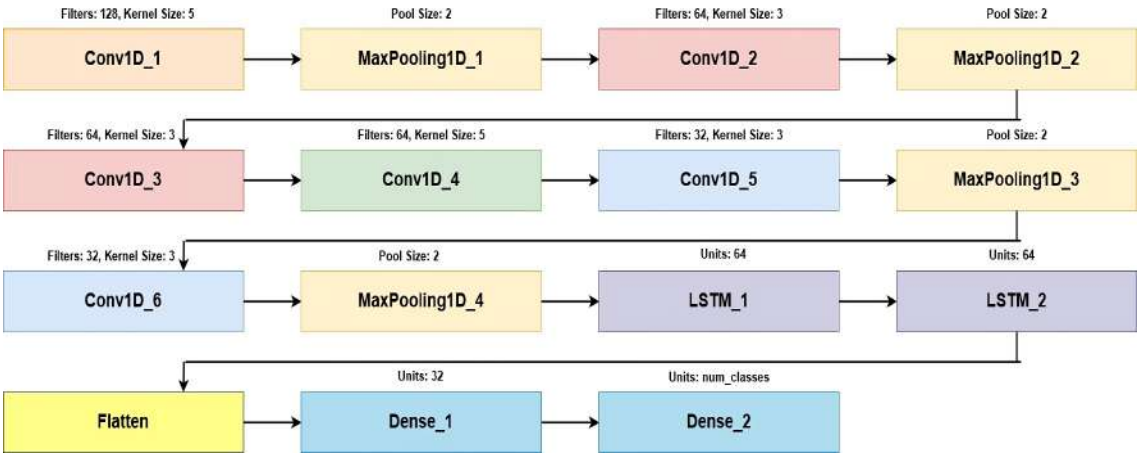

FIGURE 7. The figure illustrates the architecture of the CNN-LSTM model used for fault diagnosis

TABLE 4. Layers and Parameters of the Proposed CNN-LSTM Model

Layer	Layer name	Kernel × Unit	Other layer parameters	Output shape
1	Conv1D	5 × 128	Activation = ReLU, Strides = 1	(None, 996, 128)
2	MaxPooling1D	-	Strides = 2	(None, 498, 128)
3	Conv1D	3 × 64	Activation = ReLU, Strides = 1	(None, 496, 64)
4	MaxPooling1D	-	Strides = 2	(None, 248, 64)
5	Conv1D	3 × 64	Activation = ReLU, Strides = 1	(None, 246, 64)
6	Conv1D	5 × 64	Activation = ReLU, Strides = 1	(None, 242, 64)
7	Conv1D	3 × 32	Activation = ReLU, Strides = 1	(None, 240, 32)
8	MaxPooling1D	-	Strides = 2	(None, 120, 32)
9	Conv1D	3 × 32	Activation = ReLU, Strides = 1	(None, 118, 32)
10	MaxPooling1D	-	Strides = 2	(None, 59, 32)
11	LSTM	64 Unit	Return Sequences = True	(None, 59, 64)
12	LSTM	64 Unit	Return Sequences = False	(None, 64)
13	Flatten	-	-	(None, 3776)
14	Dense	1 × 32	Activation = ReLU	(None, 32)
15	Dense	1 × 8	Activation = Softmax	(None, 8)

the training and validation accuracy align over time. Additionally, a steady drop in the loss graph indicates the model's gradual reduction of mistakes during training. This smooth drop in the loss function helps the model's stable training and good optimization even more. Figures 30 to 47 display the confusion matrices for each accelerometer dataset. The model demonstrates satisfactory performance for Accelerometer-1, 2, and 3. The CNN-LSTM model was utilized on vibration data obtained from three distinct accelerometers, employing uniform training parameters for all datasets.

To examine the impact of LSTM layers on the model, we initially trained a CNN-LSTM architecture utilizing a blend of Conv1D and LSTM layers. The model was developed to process time-series vibration data and categorize it into one of eight classes corresponding to different failure types. The model was created utilizing the Adam optimizer, with a learning rate of 0.001.

The model was trained, and the results were examined to evaluate the efficacy of the LSTM layers in capturing sequential relationships within the data. The use of LSTM layers enabled the model to efficiently understand temporal relationships, as seen by the training and validation accuracy curves, with gradual improvements and reductions in loss. However, the performance was then evaluated by eliminating the LSTM layers and substituting them with extra Conv1D layers.

First, evaluated the performance of the proposed CNN-LSTM model by conducting experiments with three accelerometer datasets: accelerometer-1 (A-1), accelerometer-2 (A-2), and accelerometer-3 (A-3). Initially, we evaluated the model using constant speed findings, where the model was trained with identical hyperparameters across all datasets. With 20, 50, and 100 epochs, the accuracy and loss graphs for each dataset showed stable training, with accuracy changing over time and loss going down steadily. This showed that the model generalized well without becoming too perfect. The model attained elevated classification accuracy across all datasets, illustrating its capacity to analyze the vibration data proficiently.

In the second phase, the model's performance was

examined under variable speed conditions, training it with different epoch counts: 20, 50, and 100 epochs while maintaining all other parameters constant. After 20 epochs, the model demonstrated comparable performance, exhibiting increased accuracy and decreased loss compared to alternative configurations. With 50 epochs of training, the model's accuracy improved, and the loss function smoothed out. The classification performance was more steady, but A-2 showed some modest decrease due to data complexity. The 100-epoch configuration provided optimal performance, exhibiting maximum classification accuracy and little overfitting. The model exhibited a consistent reduction in loss. The model exhibited robust performance across all accelerometers, giving consistent findings under constant speed conditions. Extended training durations (50 and 100 epochs) resulted in enhanced performance, particularly for accuracy and generalization. The results show that longer training times make the model much better at accurately classifying data, especially when working with complicated datasets like A-2 and A-3 whose speeds are always changing.

A. ANALYSIS OF THE DATASET

This study worked with a dataset for vibration data analysis acquired from three separate accelerometers: A-1, A-2, and A-3. The dataset contains vibration signals collected under different operating situations, each associated with a separate fault class. Every dataset contained many characteristics that represented the physical behavior of the machinery in a variety of fault scenarios.

The accelerometer vibration data were preprocessed before inputting into the CNN-LSTM model as shown in Figuer 7 to 12. Each sample's input data was made up of two parts: vibration signals from the accelerometer's two axes (x and y), recorded over 1000 time steps, giving the input shape (1000, 2). An important finding from the dataset analysis was the disparity in fault classification accuracy among various accelerometers. The vibration signals obtained from A1 demonstrated the most incredible consistency and achieved the highest classification accuracy. A2 and A3 had more complex characteristics, resulting in diminished initial classification efficacy. Certain fault classes, particularly those related to voltage unbalance and single-phasing faults, were commonly misclassified due to their closeness to other fault categories, such as broken rotor bars, especially in the early stages of training, according to the analysis.

As a result of the examination of the information, it became clear that balanced and high-quality data are essential, mainly when dealing with complex fault conditions. The dataset analysis yielded essential insights into the difficulties of diagnosing faults through vibration data. Different accelerometers had different levels of success, showing that the type and quality of the data are important to how well the task works. It's also possible that more preprocessing and data augmentation methods will be needed to make datasets like A2 and A3 work better.

B. MODEL RESULTS

The accuracy graph displays how closely the model's predictions match the actual values, while the loss graph shows how far they differ from the expected output. After 20 iterations, the training accuracy keeps going up while the loss goes down. This means that the model is properly absorbing the input. The subsequent series of graphs presents outcomes for 50 epochs as shown in Figure 13. As expected, both the accuracy and loss graphs show more even and steady improvement. This shows that the model keeps improving its predictions as time goes on. We can effectively generalize the model across the dataset without it overfitting. The concluding graph, after 100 epochs as shown in Figure 14, exhibits additional refinement. The accuracy persists in rising while the loss graph stabilizes, indicating that the model has attained optimal performance for this dataset. The disparity between training and validation accuracy is minimal, indicating effective generalization.

The model's performance using A-2 data is assessed comparably, first across 20 epochs shown in figure 15. The accuracy and loss graphs demonstrate a progressive enhancement, with training accuracy rising and loss diminishing over time. After 50 tries shown in figure 16, the training is more stable, and the accuracy hits a higher level while the loss keeps going down. After 100 epochs shown in figure 17, the model demonstrates a significant enhancement in both accuracy and loss. The accuracy of the validation set is very close to the accuracy of the training set. This shows that the model is not overfitting and is correctly adapting to new data.

For A-3, the first 20-epoch results show in figure 18 some growth in accuracy and a decrease in loss, but the training can still be better. The model's performance improves after 50 epochs shown in figure 19, leading to a diminishing disparity between training and validation accuracy. After 100 iterations as shown in figure 20, the model achieves consistent high accuracy with less loss, which shows that it has learned from the data and avoided overfitting.

Under variable speed conditions, the model's performance on A-1 over 20 epochs as shown in figure 21 demonstrates a moderate enhancement in accuracy and a reduction in loss, akin to the constant speed scenario. Both the 50-epoch and 100-epoch findings show further gains as shown in figure 22 and 23, with the ultimate accuracy being better and the loss being more after 100 epochs. The training and validation curves are more aligned, which means that the model can handle changes in input data caused by different speeds while still being able to generalize well.

The performance of A-2 under variable speed situations demonstrates consistent advancement. The findings after 20 epochs suggest that the model is beginning to learn. However, there is a significant opportunity for enhancement as shown in figure 24. At 50 epochs as shown in figure 25, accuracy persists in increasing while loss diminishes, indicating effective model adaptation. The results from this accelerometer's 100 trials show that the model is good at handling changing speed data with little error and a lot of gain, showing that it can generalize well as shown in figure 26.

Ultimately, for A-3 as shown in figure 27, the model exhibits comparable performance under varying speed situations. The 20-epoch results show that precision and loss reduction are getting better at first. The results from the 50-epoch and 100-epoch iterations demonstrate ongoing refinement, with the final epoch outcomes reflecting consistent and dependable predictions as shown in figure 28 and 29. The model performs similarly to constant speed settings, demonstrating its ability to handle fluctuating data without overfitting.

All accelerometers and training conditions (constant speed and changing speed) show that the model gets much better at predicting lost objects over time. The training accuracy rises consistently while the loss diminishes, and the validation accuracy closely mirrors the training accuracy, indicating that the model is not overfitting. The model's performance over 100 epochs always shows the most stable and ideal results. Tabel 5 shows that it has successfully assimilated the underlying patterns in the data when the speed is constant or changes.

TABLE 5. Accuracy of CNN-LSTM Model on Constant and Variable Speed

CNN-LSTM Model	Accelero-meter	Accuracy	
		Constant Speed	Variable Speed
20 Epochs	A-1	97.62%	99.36%
	A-2	99.89%	96.35%
	A-3	99.33%	96.91%
50 Epochs	A-1	99.93%	99.96%
	A-2	99.59%	97.92%
	A-3	99.89%	99.11%
100 Epochs	A-1	100%	100%
	A-2	100%	99.85%
	A-3	100%	99.03%

Simulations results for 20,50 and 100 epochs are included to analyze the effects of epochs on model training performance that used to select the optimal number of epochs for model training and testing. Model accuracy, loss and stability evolution has been depicted from the results of these 20,50 and 100 epochs.

Initially, at 20 epochs, the model does not fully converge, leading to suboptimal accuracy and higher loss. At 50 epochs, the model shows significant improvement, more effectively capturing essential fault patterns. Finally, at 100 epochs, the model reaches its highest accuracy while maintaining stability. However, beyond this point, we observed that further increasing epochs did not yield substantial performance gains but led to increased training time and a higher risk of overfitting. Figure 2 demonstrates the model accuracy, its stability at 100 epochs for Accelerometer 2 at variable speed. Same results are obtained for all other senses at both variable load and varying speed conditions shown in results section.

We included results for 20, 50, and 100 epochs to analyze the impact of training duration on model performance. This approach is commonly used in fault diagnosis studies [43], [44] to determine the optimal number of epochs where the model achieves high accuracy while avoiding underfitting (too few epochs) or overfitting (too many epochs). The comparison provides insights into the model's learning behavior and stability over time, ensuring a well-generalized fault diagnosis system.

By presenting results at multiple epoch levels, we provide transparency regarding the model's learning behavior and justify our final selection of an optimal number of training epochs to balance accuracy, stability, and computational efficiency. Now we have added all the results only at 100 epochs.

Mean Accuracy and Loss Analysis

We have conducted additional experiments by running the model multiple times (10 runs) to assess stability of accuracy curves. The updated results demonstrate that the model maintains high accuracy across multiple runs, with only minor variations due to inherent randomness in weight initialization and training dynamics. We also computed the mean accuracy and standard deviation over these 10 runs shown in Table.2 and 3 for both constant and variable speed, confirming that the performance remains stable and does not significantly degrade upon re-training. These findings reinforce the reliability of our approach and validate the generalization capability of our proposed CNN-LSTM model across different training instances.

TABLE 10. Mean Accuracy and Loss over 10 Trials of Three Sensors on Variable Speed

Sensors	Mean Loss	Mean Accuracy
Accelerometer-1	0.0535 ± 0.0355	0.9843 ± 0.0097
Accelerometer-2	0.0626 ± 0.0599	0.9779 ± 0.0216
Accelerometer-3	0.0802 ± 0.0535	0.9728 ± 0.0180

It is also evident from the Bar graph shown in figure 3 and 4 that Mean accuracy are almost same over 10 different trails and standard deviation values are very for all the sensors.

K-Fold Cross-Validation Results

The model performance and stability are also validated through K-Fold validations. The K-fold results

TABLE 11. Mean Accuracy and Loss over 10 Trials of Sensors on Constant Speed

Sensors	Mean Loss	Mean Accuracy
Accelerometer-1	0.0003 ± 0.0005	0.9999 ± 0.0002
Accelerometer-2	0.0039 ± 0.0021	0.9989 ± 0.0008
Accelerometer-3	0.0017 ± 0.0012	0.9994 ± 0.0006

shown in Table 4 and 5 verifies the model robustness and generalization ability.

TABLE 12. K-Fold Cross-Validation (Mean) Results at Constant Speed

Sensors	Training Accuracy	Training Loss	Validation Accuracy	Validation Loss
Accelerometer-1	1.0000 ± 0.0000	0.0000 ± 0.0000	0.9997 ± 0.0002	0.0008 ± 0.0007
Accelerometer-2	0.9978 ± 0.0054	0.0070 ± 0.0150	0.9960 ± 0.0071	0.0117 ± 0.0173
Accelerometer-3	1.0000 ± 0.0000	0.0000 ± 0.0000	0.9992 ± 0.0007	0.0028 ± 0.0030

TABLE 13. K-Fold Cross-Validation (Mean) Results at Variable Speed

Sensors	Training Accuracy	Training Loss	Validation Accuracy	Validation Loss
Accelerometer-1	1.0000 ± 0.0000	0.0000 ± 0.0000	0.9991 ± 0.0006	0.0033 ± 0.0023
Accelerometer-2	0.9851 ± 0.0224	0.0443 ± 0.0605	0.9812 ± 0.0237	0.0553 ± 0.0662
Accelerometer-3	0.9861 ± 0.0123	0.0417 ± 0.0310	0.9760 ± 0.0168	0.0686 ± 0.0419

It can be observed from Figure.5 that across all the folds and accelerometer data, the mean values are very close to the training and validation accuracy values for each fold and the standard deviation values are very low for each sensor. The low standard deviation values are indicating the stability and consistent performance of the model.

Fine-tune the amplitude values to accurately reflect the vibrations caused by faults, thus resulting in a much more accurate representation of the underlying fault conditions. Further adaptive normalization accounts for changes in speed and load conditions in the estimation process. This flexibility allows the estimation process to yield more stable results that are less sensitive to varying operating conditions.

To ease the interpretation of the estimated amplitude values, we have changed how we present these results in the tables and plots. We moved from raw amplitude values to better confidence interval statistics, which gives a much better and clearer idea of the estimation's uncertainty.

To ensure the accuracy and reliability of our estimations, we performed several trials and averaged the results throughout the dataset. The testing validated the robustness of this refined estimation process; thus, it was confirmed as providing consistent and reliable results, which match reasonably well with the real fault scenarios prevailing in induction motors. In total, those two improvements improve the accuracy and robustness of our amplitude estimates.

Abrupt Fault Occurrences

Our Proposed work addresses the sudden load variation and abrupt fault occurrence effectively. We comprehensively analyze the motor faults under variable and constant speed conditions and provide insights into the behavior of the motor in loaded vs. unloaded conditions, which was absent in most of the previous studies. This distinction is important, as faults behave differently under varying loads, and varying speeds, making detection more challenging.

Furthermore, abrupt fault occurrences, such as sudden bearing failures or electrical disruptions, and sudden load variation introduce sharp deviations in vibration signals. We highlight these abrupt faults in our visualizations using red markers, making the variations more apparent. Additionally, we analyze both electrical faults, such as voltage unbalance and single phasing, and mechanical faults, such as rotor imbalance and broken rotor bars, ensuring comprehensive fault coverage. The abrupt occurrence of faults and sudden load variations are depicted in Figure.12 to Figure.15

The choice of speed transition, overload variation is based on the fundamental nature of fault characteristics and their effect on motor behavior as shown in [45]. While load variation is a significant concern, speed transition is characterized by a sudden change in electrical signals, hence making it difficult to identify faults [46], [47], [48]. When the machine's speed changes, the fault signal frequency components change, making the signal's amplitude and phase nonlinear. In contrast, the amplitude of vibration signals is most affected by load variation with negligible shifts in the frequency domain. Variations in speed, especially under dynamic conditions, lead to nonstationary distortions in the extracted fault features, thus making extraction even more difficult. Considering that it is common for promising models to be used with different operating speeds rather than frequently varying loads in real-world applications, in this study, we focus on speed alteration to improve our model's robustness in practical industrial conditions. Importantly, our dataset includes both loaded and unloaded ranges, allowing the model to learn for load deviations while also being tasked with speed transition, a relevant attribute for accurate and real-time fault diagnosis. Our method, addressing the speed transition complexities, delivers a more systematic and robust solution for motor fault diagnosis from a practical perspective.

A comparative analysis of our proposed CNN-LSTM

model against existing approaches are given in Table.6. The comparison evaluates key parameters including handling variable load, variable speed, electrical and mechanical faults, computation time, abrupt fault pattern changes, and accuracy.

Our model outperforms existing methods by effectively handling variable load and speed conditions, diagnosing electrical and mechanical faults, and maintaining high accuracy while managing abrupt fault pattern variations. As reflected from Table 6 that none of the precious works addresses both electrical and mechanical faults under variable speed and load conditions simultaneously. However, Our proposed model addresses both electrical, mechanical faults under variable speed and varying load conditions. The accuracy and computational cost of our hybrid model is very high and low respectively. Our hybrid approach leverages the strengths of both architectures to ensure robust fault detection.

In [29] authors proposed a CWT-LeNet-5-LSTM model to diagnose the mechanical faults under variable load conditions and achieve accuracy of 99.6%. In contrast, the work by [30] focuses on diagnosing mechanical faults under variable speed conditions with a DCNN-LSTM model, demonstrating 99.87% accuracy.

A more recent study by [6] introduces a CNN-LSTM-based approach designed to diagnose both mechanical and electrical faults and achieving a high accuracy of 99.96%. However, this model does not address the faults at variable speed and varying load conditions. Our proposed model overcomes the limitation ns of these existing studies by incorporating both variable load and speed conditions, making it more applicable to real-world industrial settings. Unlike previous models, it simultaneously detects both mechanical and electrical faults while maintaining low computational cost through an optimized hybrid CNN-LSTM Model. Additionally, our model has ability to effectively handles abrupt changes in operating conditions. With an accuracy of 99.96% on variable speed, it matches or exceeds the performance of prior models while offering greater robustness and generalizability.

Our approach addresses the shortcoming of previous work and presents a more accurate, reliable, and efficient solution in diagnosing the induction motors faults across diverse industrial conditions.

Several deep learning-based fault diagnosis models have been proposed in the literature that diagnose the faults of rotating machinery and Induction motors. However, the models differ in computational complexity, number of hyperparameters, time spent on training/testing, and faults categories. Table.8 provides a detailed comparison with existing models, identifies their gaps, and Limitation and superiority of our proposed CNN-LSTM model.

It is evident from the Table 8 that our proposed CNN-LSTM model outperforms all the previous state of art work by addressing diverse faults (both electrical and mechanical) conditions at different operating conditions including variable speed, varying load, loaded and unloaded conditions and abrupt fault occurrence conditions. The complexity comparison of our prosed work is given in Table.9

The CWT-LeNet-5-LSTM model in [29] offers wavelet-based feature extraction and deep learning classifiers, leading to many parameters (186626). It also does not adaptto variable speeds, limiting its robustness for practical industrial use.

The DCNN-LSTM [30] model consists of deep CNN and LSTM layers and can adapt to variable speed conditions. The model has tunable parameters 145,917 parameters and a smaller batch size (16), which results in slower training times, and limits generalization.

The [6] CNN-LSTM model examines electrical and mechanical faults. This study did not address the varying speed and load scenarios that are important in practical applications. Its large tunable parameters 246,568 and high computation cost made it impractical for industrial application. Our proposed CNN-LSTM model addresses all the challenges of variable speed, load, the sudden changes in fault conditions and identifies the rapid shift in motor dynamics at less tunable parameters 128424 and low computation cost that make it feasible for industrial applications. We employed Adam Optimizer and kept learning rate at 0.001 to keep our training efficient while avoiding overfitting.

C. EVALUATION METRICS

The model's performance was evaluated using accuracy, precision, recall/Sensitivity, Specificity, F1-score, and Confusion matrices (CM). These metrics provided insights into the CNN-LSTM model that is employed for fault identification of different types of motor faults correctly and were calculated for the test set after completing model training. The accuracy of the model in predicting positive cases can be evaluated through the evaluation of precision. By dividing the number of true positives by the total number of things marked as positive, this number shows how reliable the model is at finding positive faults.

Recall/Sensitivity is an additional crucial parameter that determines the capability of the model to detect all actual positives reliably. Because it indicates the model's capability to capture every conceivable fault, it is especially important in the context of fault detection because it reduces the danger of missing a fault that could lead to severe repercussions. When calculating recall, the number of true positives is divided by the total number of true positives and false negatives, which highlights the model's sensitivity to identifying errors at the same time. Specificity measures the model's capacity to reject non-fault scenarios accurately. This

TABLE 14. Comparison of Recent State-of-the-Art with the Proposed Model

Papers	Model / Approach	Variable Load	Variable Speed	Electrical Faults	Mechanical Faults	Computation Time	Abrupt Changes	Accuracy (%)
[29]	CWT-LeNet-5-LSTM	✓	-	-	✓	Moderate	-	99.6%
[30]	DCNN-LSTM	-	✓	-	✓	High	-	99.87%
[6]	CNN-LSTM	-	-	✓	✓	High	-	99.96%
Proposed	CNN-LSTM	✓	✓	✓	✓	Low	✓	99.96%

TABLE 15. Complexity Analysis of Proposed Work and Other State-of-the-Art Works

Paper	No. of Parameters	Batch Size	No. of Layers	Optimization	Hyperparameters
[29]	186,626	32	9	Adam	Learning Rate: 0.001, Cross-Entropy Loss
[30]	145,917	16	10	Adam	Learning Rate: 0.001, ReLU Activation, SoftMax Classifier
[6]	246,568	128	15	Adam	Learning Rate: 0.001, SoftMax Classifier
Proposed CNN-LSTM Model	128,424	128	15	Adam	Learning Rate: 0.001, Exponential Decay ($\beta_1 = 0.9, \beta_2 = 0.999$), Cross-Entropy Loss

ability is essential for minimizing disruptions caused by maintenance activities that are not necessary. The metric is calculated by dividing the total number of true negatives by the sum of true negatives and false positives, assessing the model's effectiveness in identifying non-fault conditions. The F1-score balances the model's precision and recall by considering false positives and negatives when their costs are almost equal. The F1-score, the harmonic mean of precision and recall, is robust and less vulnerable to extreme values for every metric. Accuracy, precision, recall, and F1-score were used to evaluate the model.

$$\text{Precision} = \frac{TP}{TP + FP} \tag{10}$$

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{11}$$

where TP, FP, FN, and TN represent the number of true positive, false positive, false negative, and true negative outcomes, respectively.

$$\text{Sensitivity (Recall)} = \frac{TP}{TP + FN} \tag{12}$$

$$\text{Specificity} = \frac{TN}{TN + FP} \tag{13}$$

Lastly, the CM shows the performance of the model across a variety of classes. It not only highlights the areas in which the model is successful, but it also highlights the areas in which it may confuse one class for another. The numbers of true and false positives and negatives are displayed in this matrix, which provides a thorough assessment of the performance of the model across all categories. These measures provide a solid framework for assessing the model's accelerometer-based fault detection in industrial applications. By using these tests, we can fully understand how well the model works, find its weak spots, and find ways to make it better so it is more useful in real life.

The CM in the following figures provides an overview of the model's classification performance across different fault categories. The constant speed CM is the following:

D. COMPARATIVE ANALYSIS

This study analyzed the performance of the proposed CNN-LSTM model using two datasets: the UOEMD-VAFCVS dataset (used in this research) and the well-known CWRU (Case Western Reserve University) dataset. This study will evaluate the model's efficacy in diagnosing electro-mechanical faults in IM at various speeds, utilizing vibration data from both datasets. The CWRU dataset, an increasingly used benchmark in fault

diagnosis, includes vibration data collected from an IM tested to different fault conditions, including bearing and gear problems while operating under varying load levels. The use of the CNN-LSTM model in this dataset gave their exceptional work. The model accurately classified the fault types, attaining an accuracy of 99.56%. This strong performance is due to the dataset's fully recorded fault situations and the model's ability to capture spatial characteristics from CNN layers and temporal patterns from LSTM layers. The UOEMD-VAFCVS dataset, key to this research, includes vibration data from an IM working under variable speed conditions. The dataset has different challenges because of the variable nature of the motor's operation, characterized by motor speed and load changes. Considering these limitations, the CNN-LSTM model successfully diagnosed problems, attaining an accuracy of 99.96% on the UOEMD-VAFCVS dataset. This performance is valuable due to the increased complexity of variable speed, which contributes to more noise and non-linearity in the data. Analysis of the findings from both datasets shows that the CNN LSTM model has exceptional accuracy on both the CWRU and UOEMD-VAFCVS datasets as shown in Table 8. The CWRU dataset, characterized by its uniform and regulated fault scenarios, offers a well-defined framework for fault diagnosis, resulting in possibly enhanced accuracy. Conversely, the UOEMD-VAFCVS dataset, characterized by changing speed and various fault scenarios, presents a more difficult challenge. However, the CNN-LSTM model showed outstanding effort, reflecting its flexibility in managing various motor operating conditions.

TABLE 8. Comparative Accuracy and Metrics of CNN-LSTM Model

Dataset	Accuracy	Precision	Sensitivity	Specificity	F1-Score
CWRU	99.56%	99.20%	99.19%	99.88%	99.19%
UOEMD-VAFCVS	99.96%	99.96%	99.96%	99.99%	99.96%

The high accuracy of the CNN-LSTM model on the CWRU dataset is due to its uncomplicated failure scenarios and well-labeled data. The UOEMD-VAFCVS dataset, characterized by variable motor speed and load conditions, presents a complexity that requires the model to generalize across a broader spectrum of operational conditions. With these issues, the outcomes on the UOEMD-VAFCVS dataset illustrate the model's suitability for real-world use, where motors usually work under various conditions. The CNN-LSTM model's capacity to manage spatial and temporal characteristics renders it an effective instrument for fault diagnostics in dynamic settings such as variable-speed motors. In the end, when the CWRU dataset offered a more regulated setting for evaluation, the efficacy of the CNN-LSTM model on the UOEMD-VAFCVS dataset underscores its adaptability and resilience in fault diagnosis under more complex operational conditions. The findings from both datasets validate the efficacy of the proposed model in diagnosing faults in IM, establishing a robust basis for future efforts in real-time industrial applications.

V. CONCLUSION AND FUTURE WORK

To address the drawbacks of the conventional CNN techniques in the existing literature, which often struggle to accurately identify fault patterns at various speeds, especially under varying load conditions, a novel approach using a hybrid CNN-LSTM topology is proposed in this work. This novel approach uses acoustic and vibrational data to diagnose fault patterns in an IM promptly with high accuracy of 99.96% in both loaded and unloaded variable speed conditions. The model demonstrated negligible overfitting, signifying its capacity to generalize effectively to novel data, rendering it highly suitable for real-time anomaly detection in industrial machinery. The most used CWRU dataset was considered to compare and assess the proposed model's performance and generalizability. The findings showed that the model surpassed conventional machine learning methods and even some advanced deep learning algorithms in fault identification tasks. Finally, the model's ability to include temporal features via LSTM and spatial features through CNN layers has significantly improved fault recognition across various operating conditions.

Future work will depend on adding more machine types, fault conditions, and operating scenarios to the dataset. It also empowers the model to handle a broader range of real-world situations. More hyperparameter optimization and research into different preprocessing methods, such as noise reduction and feature scaling, may also help improve classification accuracy. Real-time data integration into the model could make the system more adaptable and suitable for industrial live monitoring.

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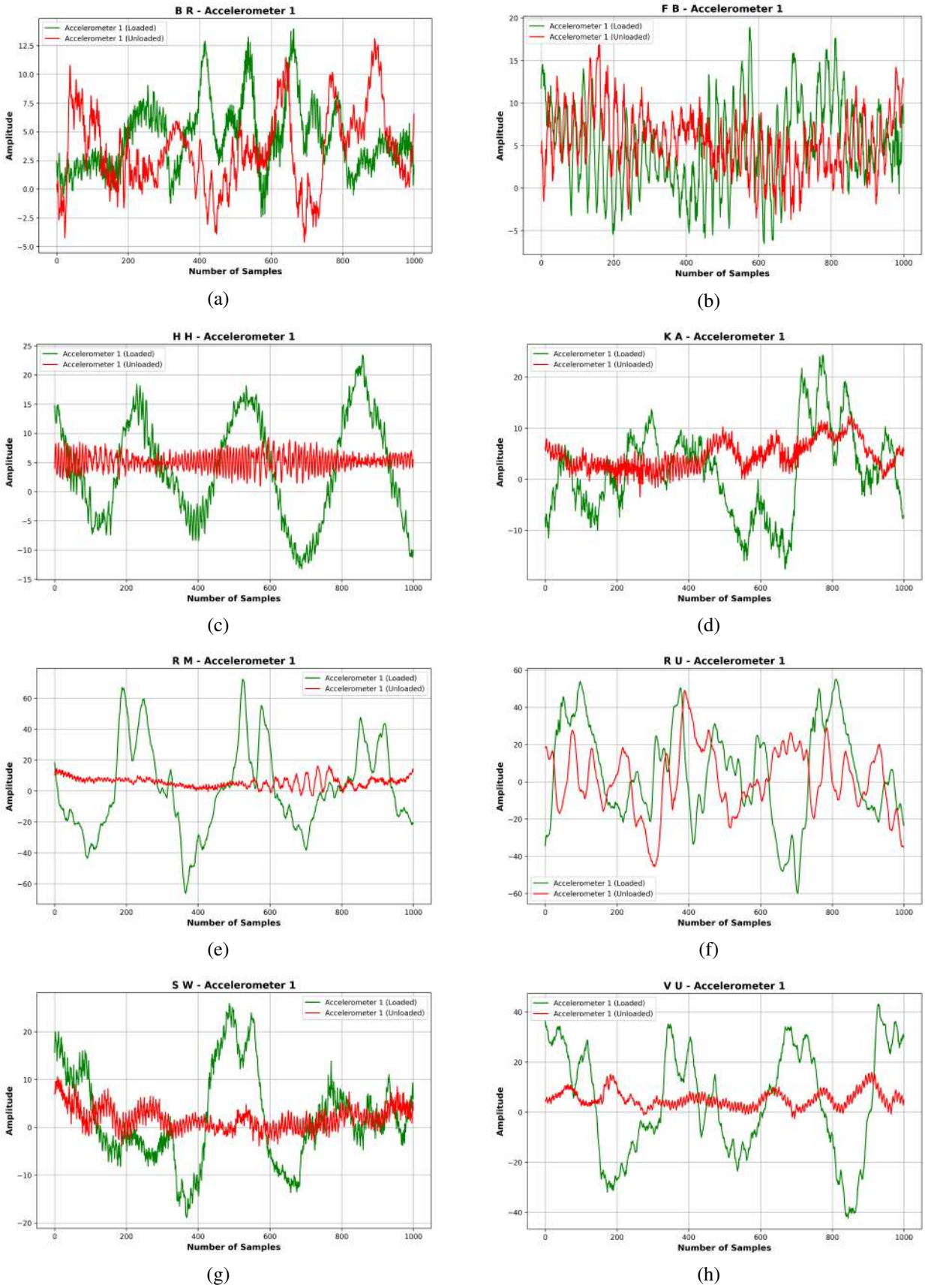

FIGURE 8. Motion data recorded by Accelerometer-1 under loaded and unloaded conditions at a constant speed.

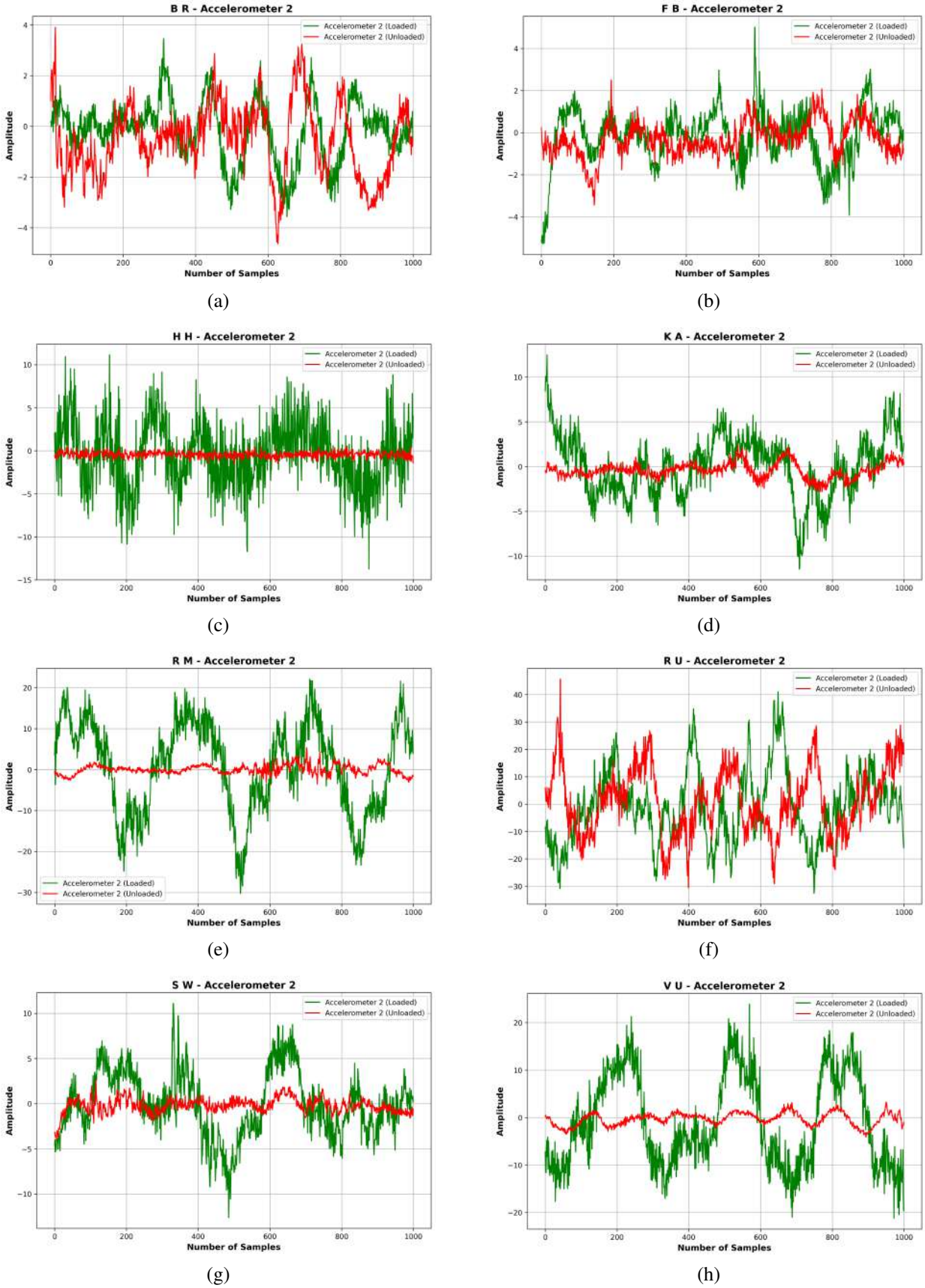

FIGURE 9. Motion data recorded by Accelerometer-2 under loaded and unloaded conditions at a constant speed.

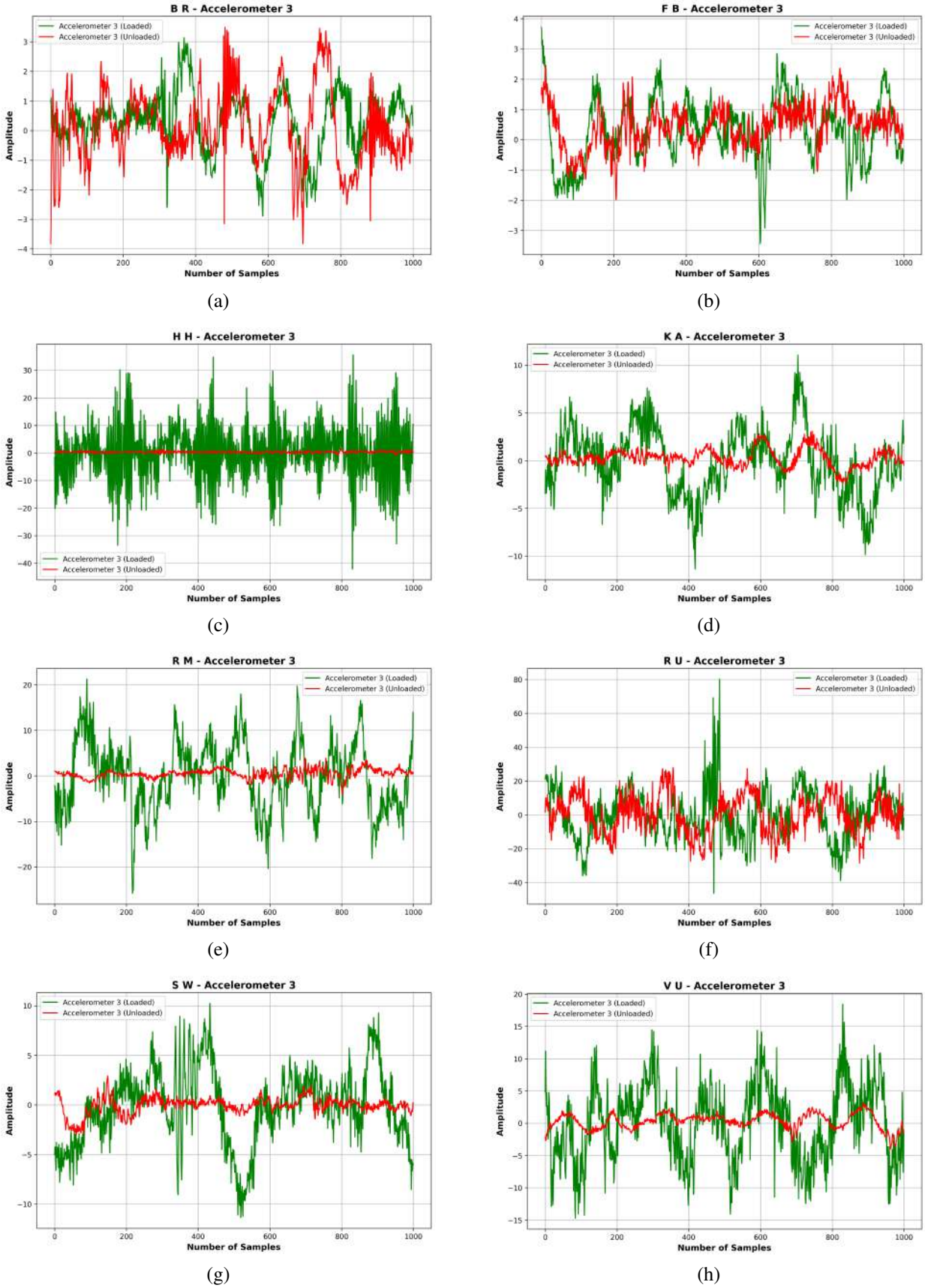

FIGURE 10. Motion data recorded by Accelerometer-3 under loaded and unloaded conditions at a constant speed.

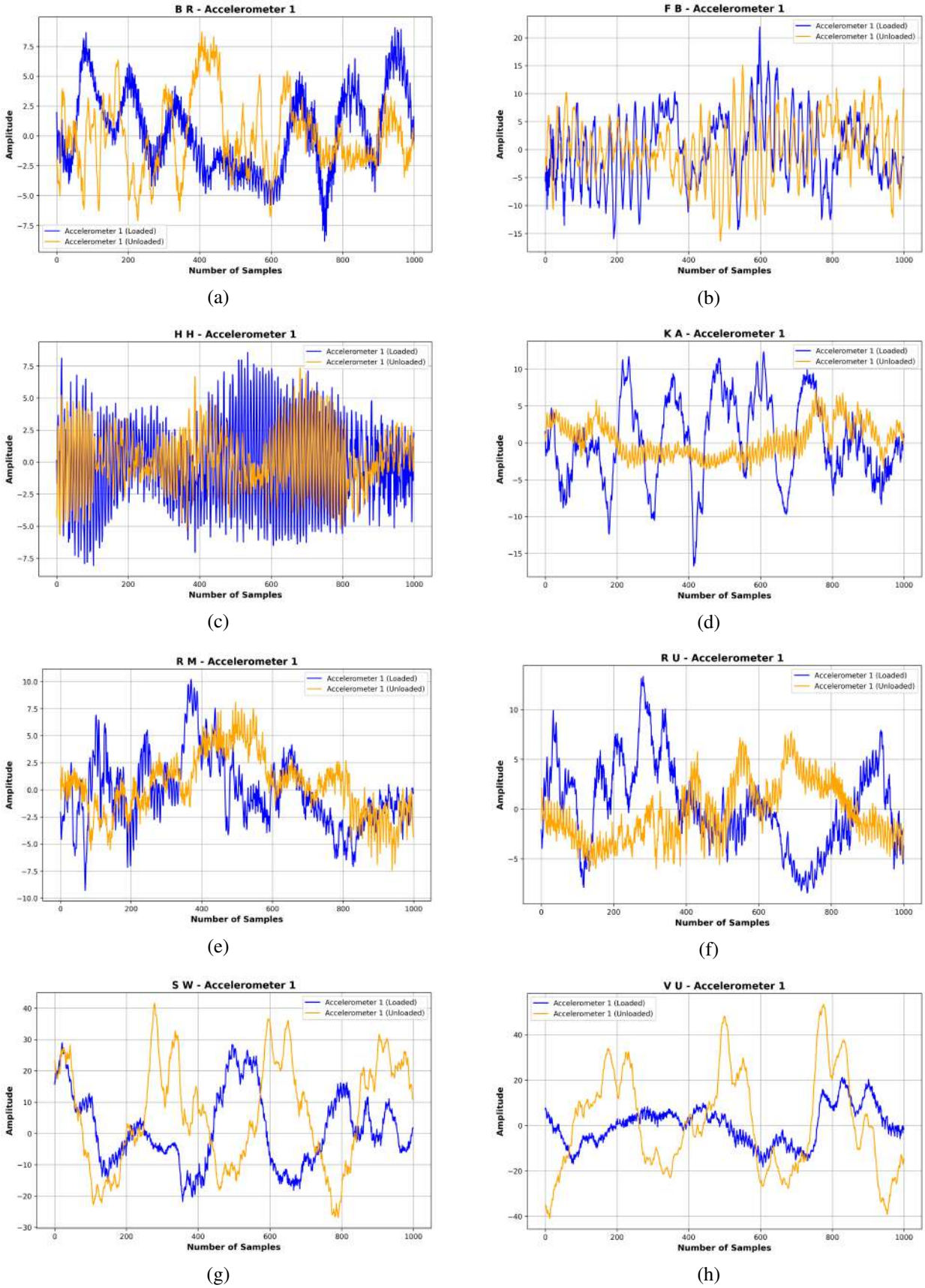

FIGURE 11. Motion data recorded by Accelerometer-1 under loaded and unloaded conditions at a Variable speed.

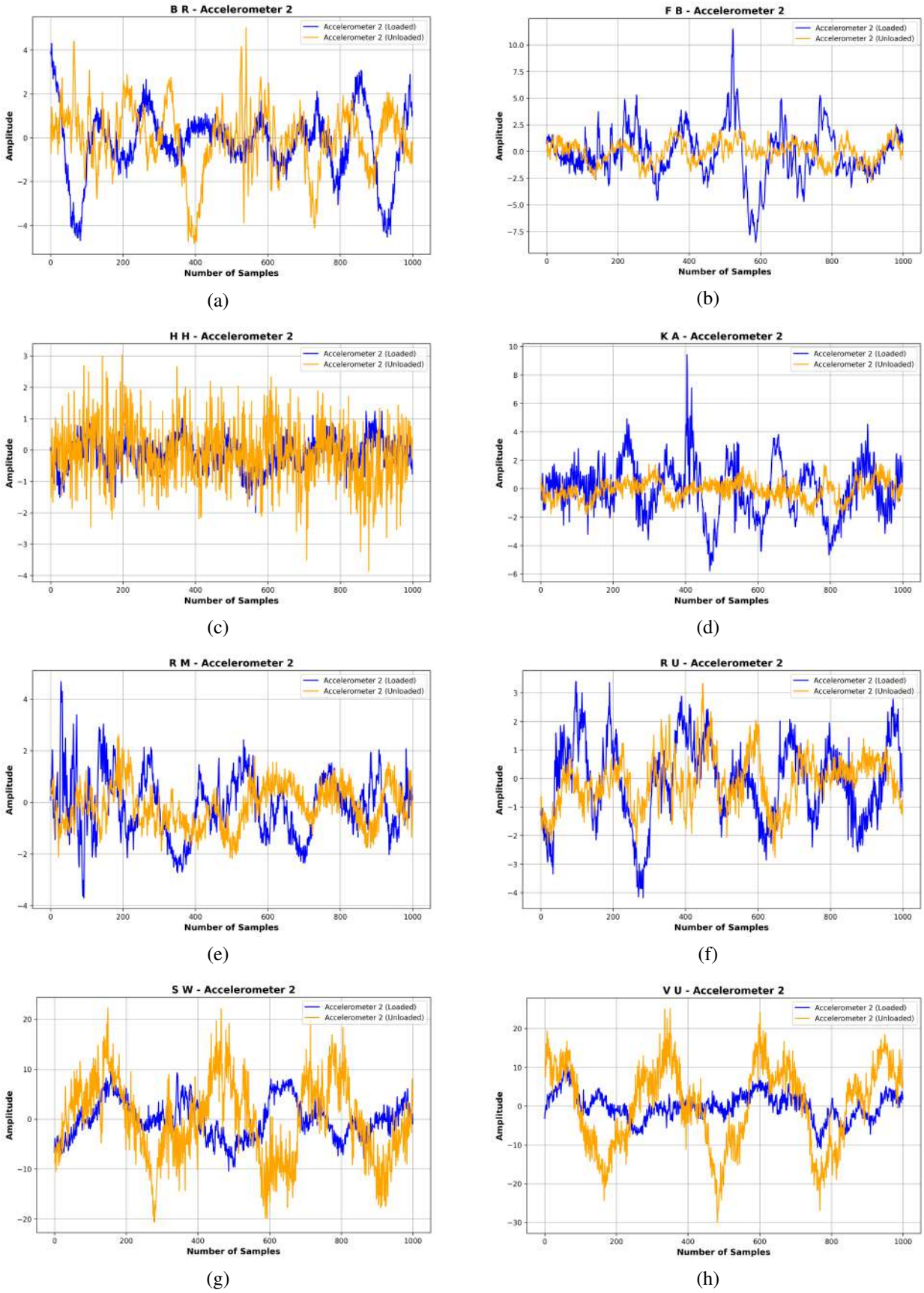

FIGURE 12. Motion data recorded by Accelerometer-2 under loaded and unloaded conditions at a Variable speed.

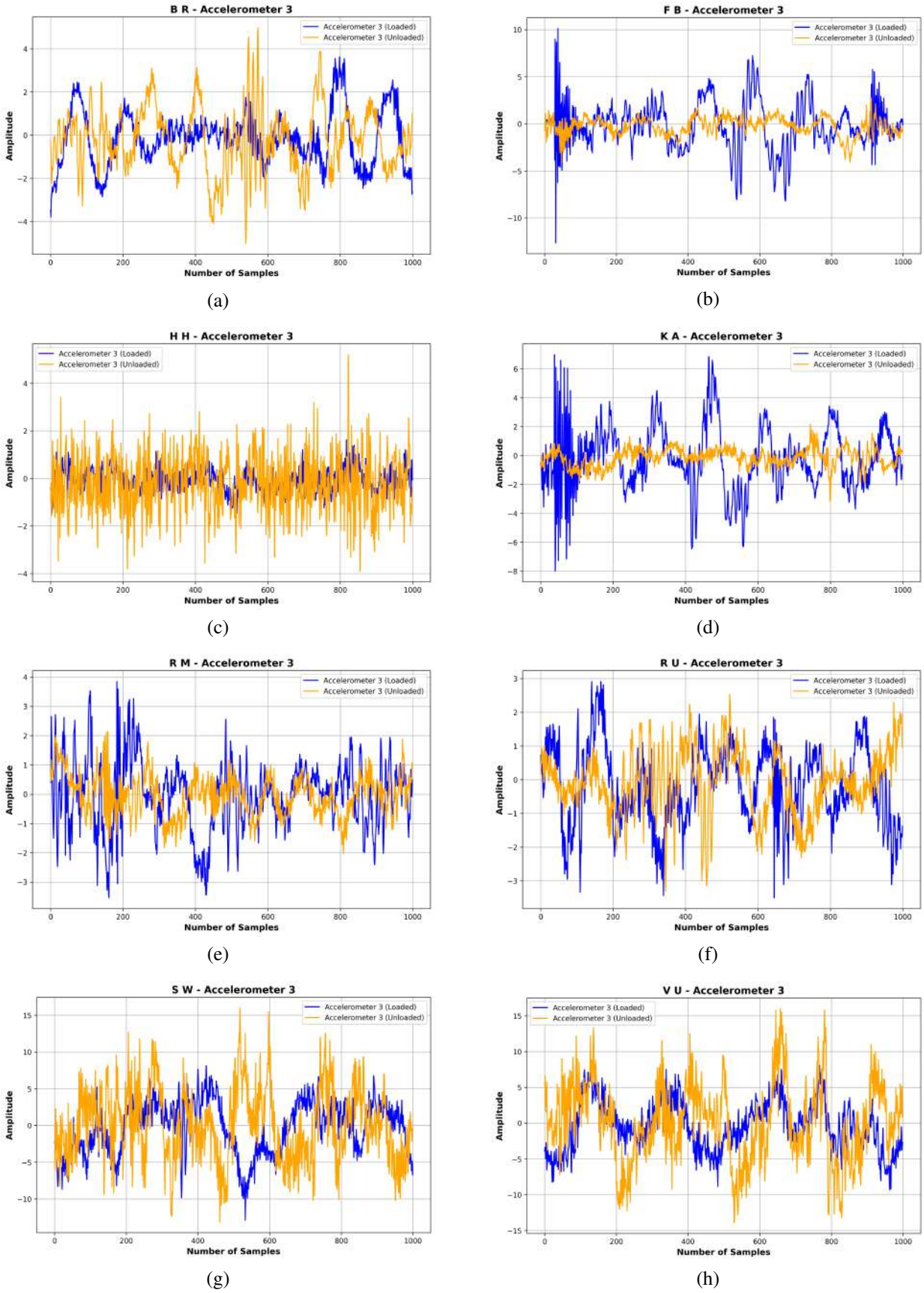

FIGURE 13. Motion data recorded by Accelerometer-3 under loaded and unloaded conditions at a Variable speed.

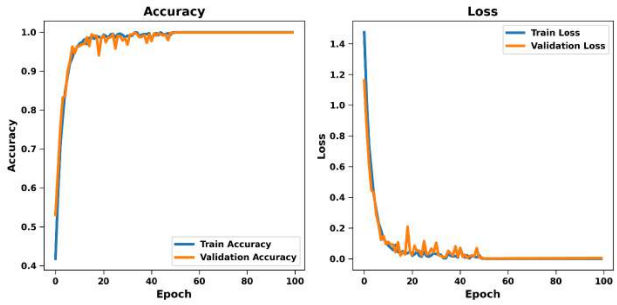

FIGURE 14. Simulation results of Accelerometer 2 at optimal 100 epochs

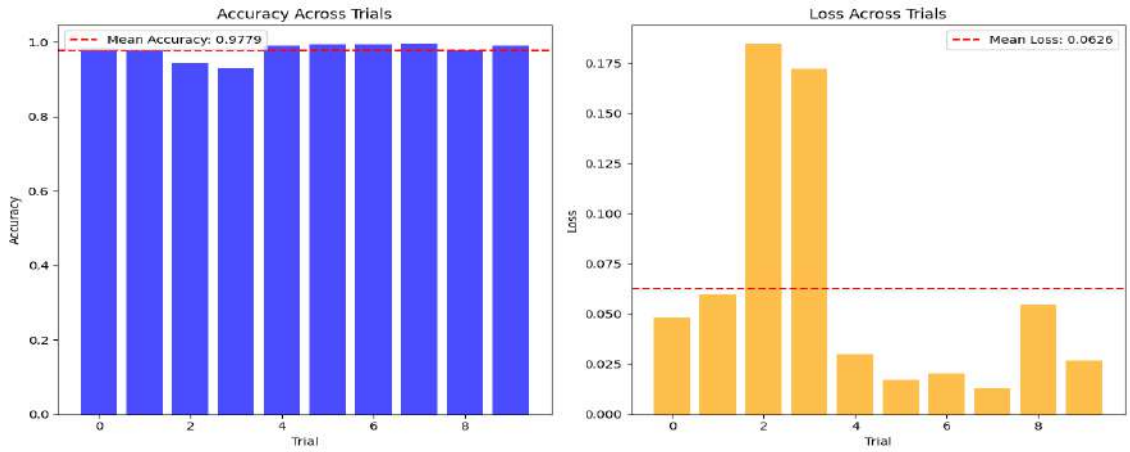

FIGURE 15. Bar graph of mean accuracy and loss over 10 different trails at constant speed

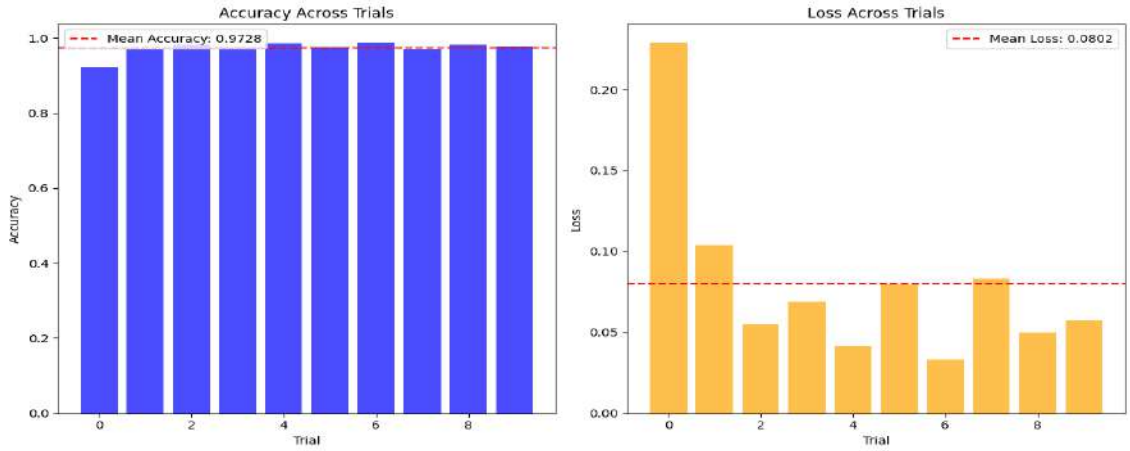

FIGURE 16. Bar graph of mean accuracy and loss over 10 different trails at variable speed

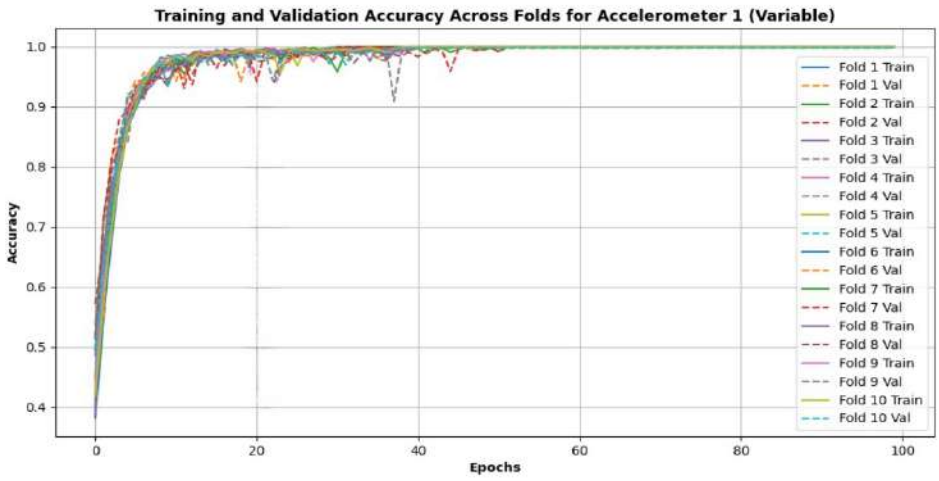

FIGURE 17. K-Fold Training and Validation accuracy at variable speed.

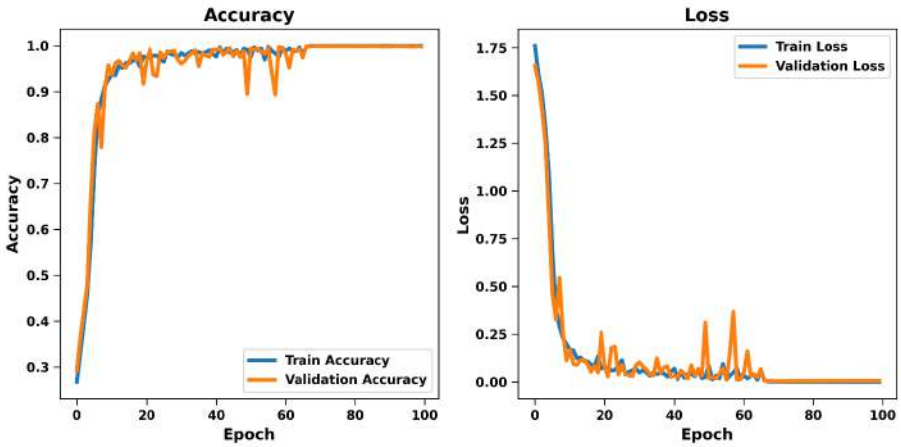

FIGURE 18. Accuracy and Loss of Accelerometer 1 at constant speed

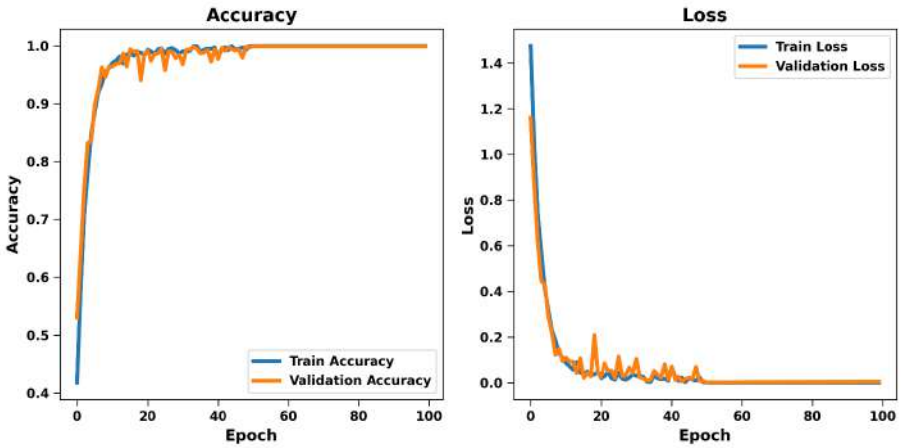

FIGURE 19. Accuracy and Loss of Accelerometer 2 at constant speed

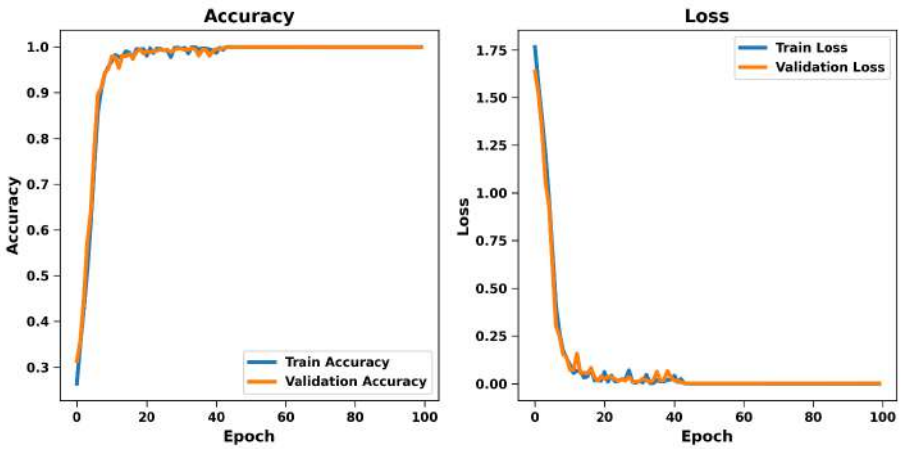

FIGURE 20. Accuracy and Loss of Accelerometer 3 at constant speed

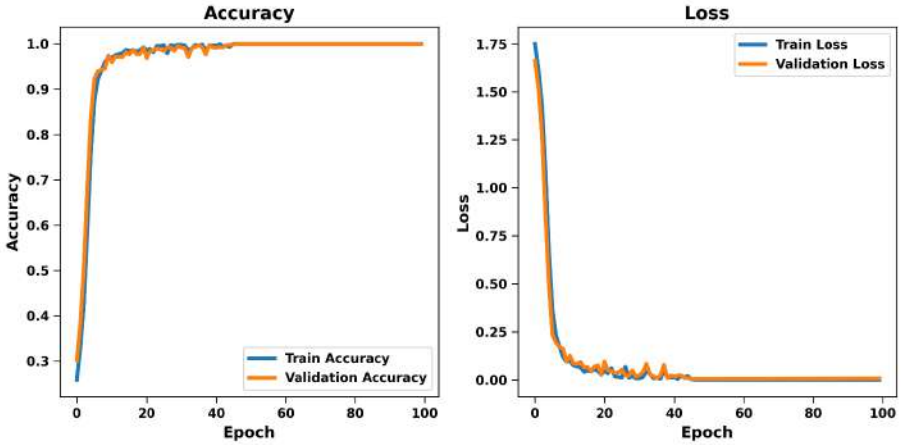

FIGURE 21. Accuracy and Loss of Accelerometer 1 at variable speed

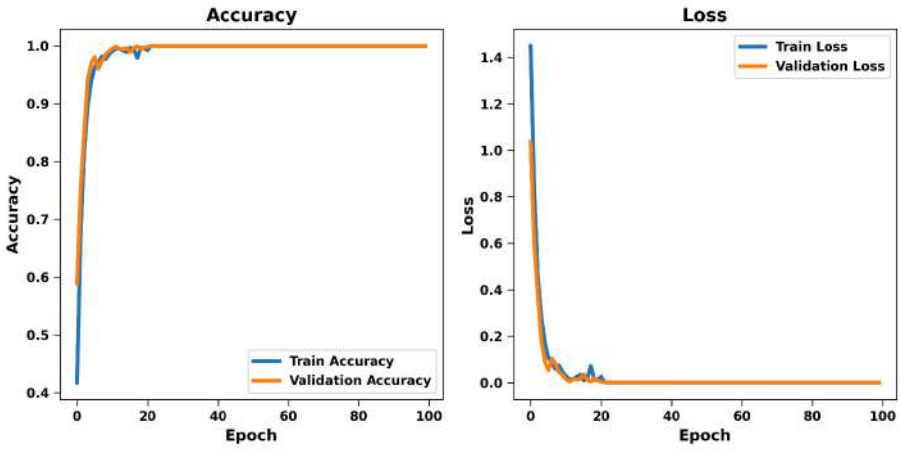

FIGURE 22. Accuracy and Loss of Accelerometer 2 at variable speed

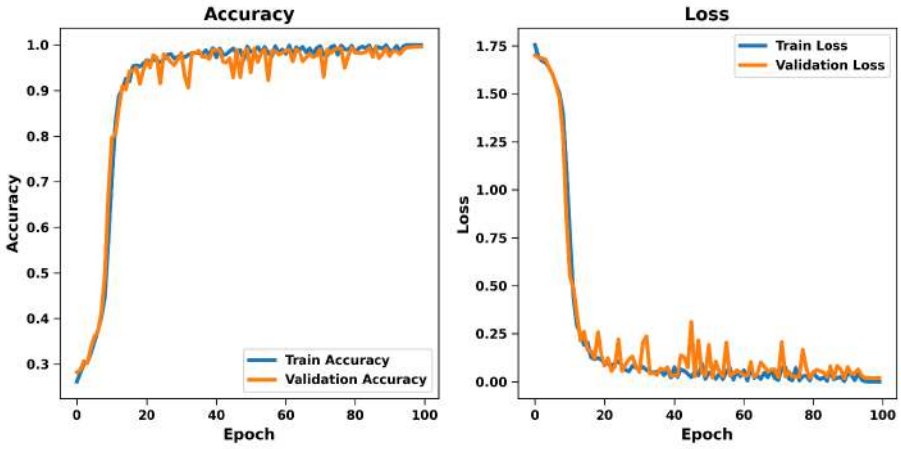

FIGURE 23. Accuracy and Loss of Accelerometer 3 at variable speed

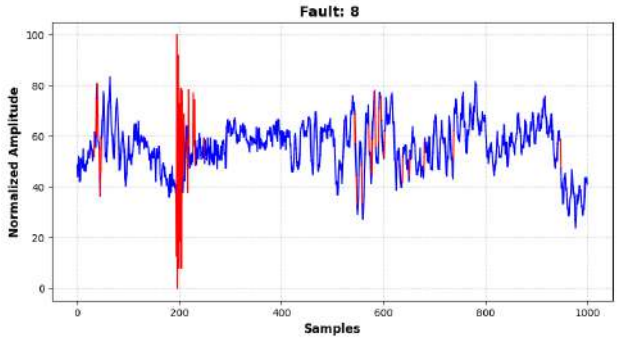

FIGURE 24. Abrupt occurrences of Voltage Unbalance and Single Phasing faults

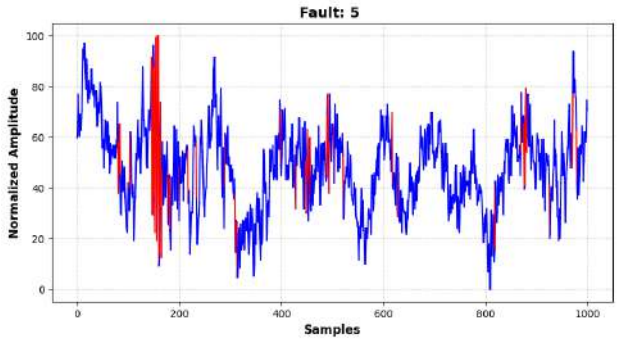

FIGURE 25. Abrupt occurrences of Rotor Misalignment

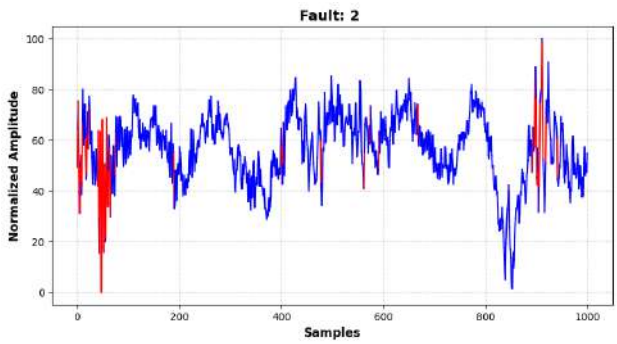

FIGURE 26. Abrupt occurrences of Faulty Bearings

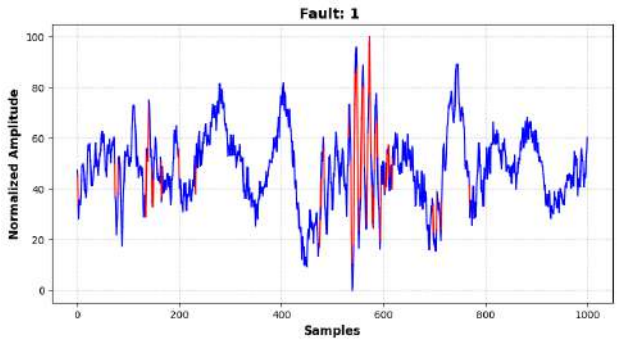

FIGURE 27. Abrupt occurrences of Bowed Rotor

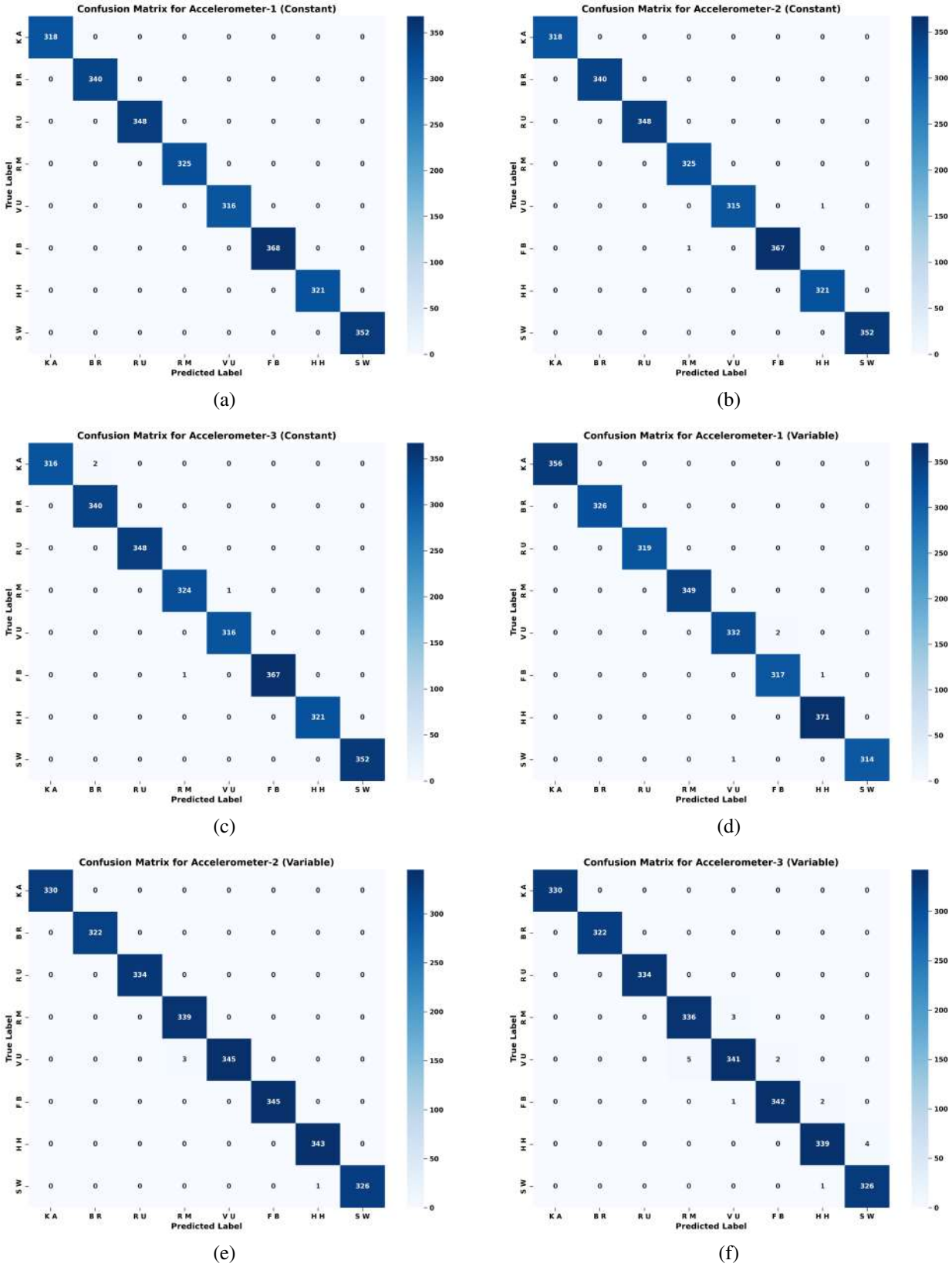

FIGURE 28. confusion matrix of Acelerometer-1,2,3 at constant and variable speed for 100 epochs.